

# Techno-economic analysis of a grid/fuel cell/PV/electrolyzer system for hydrogen and electricity production in the countries of African and Malagasy council for higher education (CAMES)

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## ABSTRACT

The current electricity challenges in Africa have spurred the interest of governments in incorporating renewable energy sources into the energy mix. The use of hybrid solar energy systems has become a practical choice for electrification. The novelty of the paper is threefold: (i) The Grid/Fuel Cell/PV/Electrolyzer hybrid system is modeled, simulated, and optimized in for some communities of the nineteen countries of the African and Malagasy Council for Higher Education (CAMES) for the very first time; (ii) The leveled costs of hydrogen and energy were evaluated and compared; (iii) PV, fuel cell, grid purchases, and grid sales power productions were determined and compared. The study findings showed a leveled cost of energy varying between US\$ 0.238/kWh (in Madagascar) and US\$ 0.344/kWh (in Gabon). Subsequently, among the West, Central, and East African countries of CAMES, Mali, Chad, and Burundi were the countries with the smallest LCOE respectively. Also, the leveled cost of hydrogen ranged between US\$ 1.84/kg (in Madagascar) and US\$ 2.17/kg (in Gabon and Equatorial Guinea). The computation of the net present cost indicated a variation between 183,536 US\$ (in Madagascar) and 216,115 US\$ (in Gabon). The PV, fuel cell, grid purchases, and grid sales power productions were between 34,044 kWh/year (in Gabon) and 53,176 kWh/year (in Mali) for PV productions; between 43,526 kWh/year (in the Democratic Republic of Congo) and 43,784 kWh/year (in Chad) for fuel cell productions; between 29,976 kWh/year (in Senegal) and 30,700 kWh/year (in the Congo Republic) for grid purchases power productions; and between 11,604 kWh/year (in Gabon) and 29,608 kWh/year (in Mali) for grid sales productions respectively. It was concluded that the PV power production was the highest power generated by the hybrid system when compared with fuel cell, grid sales, and grid purchases power productions. Concerning the PV penetration, it varied between 54.2 % (in Gabon) and 84.7 % (in Mali). The present work is aimed at decision-makers so that they become aware of the need to urgently develop the renewable energy sector in the countries for a successful energy transition in Africa.

## 1. Introduction

The African and Malagasy council for Higher Education (CAMES) is a regional institution which organizes the coordination of higher education policies and research in nineteen countries (Benin, Burkina Faso, Burundi, Cameroon, Central African Republic, Congo Republic, Ivory Coast, Gabon, Guinea, Guinea Bissau, Equatorial Guinea, Madagascar, Mali, Niger, Democratic Republic of Congo, Rwanda, Senegal, Chad and

Togo) [1]. The role of CAMES is to support a sustainable development of science, innovation and technology in Africa [1]. The countries of CAMES face various energy challenges [2]: a poor access to electrical energy; unreliable electrical energy networks; and high dependence on fossil fuels. However, a very strong potential in renewable energies is observed but under-exploited [2,3]. Thus while the planning of a transition towards renewable and sustainable energy systems is an ongoing and pertinent step, so is the development of the technologies and approaches that are required for the transition [4]. Therefore, the present

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| Nomenclature       |                                                                          |                           |                                                                                           |
|--------------------|--------------------------------------------------------------------------|---------------------------|-------------------------------------------------------------------------------------------|
| $\delta$ :         | Solar declination [Degrees]                                              | $R_b$ :                   | Ratio of beam radiation on the tilted surface to beam radiation on the horizontal surface |
| $n$ :              | Day of the year                                                          | $A_i$ :                   | Anisotropy index                                                                          |
| $\omega$ :         | Hour angle [Degrees]                                                     | $f$ :                     | Factor used to account for horizon brightening                                            |
| $t_c$ :            | Civil time [Hours]                                                       | $\rho_g$ :                | Ground reflectance [%]                                                                    |
| $t_s$ :            | Solar time [Hours]                                                       | NPC:                      | Net Present Cost [US\$]                                                                   |
| $\lambda$ :        | Longitude [Degrees]                                                      | LCOH:                     | Levelized cost of Hydrogen [US\$/kg]                                                      |
| $Z_c$ :            | Time zone [Hours]                                                        | LCOE:                     | Levelized cost of energy [US\$/kWh]                                                       |
| $\theta$ :         | Angle of incidence [Degrees]                                             | CRF:                      | Capital recovery factor                                                                   |
| $\beta$ :          | Slope of the surface of the PV array [Degrees]                           | $C_{annualized}$ :        | Annualized cost [US\$/year]                                                               |
| $\gamma$ :         | Azimuth of the surface of the PV array [Degrees]                         | $i$ :                     | Annual real discount rate [%]                                                             |
| $\varphi$ :        | Latitude $\delta$ [Degrees]                                              | $R_{proj}$ :              | Project lifetime [years]                                                                  |
| $\theta_z$ :       | Zenith angle [Degrees]                                                   | $C_{annualized, total}$ : | Total annualized cost of the system [US\$/year]                                           |
| $G_{on}$ :         | Extraterrestrial normal irradiance [kWh/m <sup>2</sup> /day]             | $E_{served}$ :            | Total electrical load served [kWh/year]                                                   |
| $G_0$ :            | Extraterrestrial horizontal irradiance [kWh/m <sup>2</sup> /day]         | $\nu_{elec}$ :            | Value of electricity                                                                      |
| $\overline{G_0}$ : | Average extraterrestrial horizontal irradiance [kWh/m <sup>2</sup> /day] | $E_{prim}$ :              | Primary electrical load [kWh/year]                                                        |
| $\omega_1$ :       | Hour angle at the beginning of a time step [Degrees]                     | $E_{Def}$ :               | Deferrable load [kWh/year]                                                                |
| $\omega_2$ :       | Hour angle at the end of a time step [Degrees]                           | $E_{grid, sales}$ :       | Total energy sold to the grid [kWh/year]                                                  |
| $k_T$ :            | Clearness index                                                          | $M_{Hydrogen}$ :          | Total hydrogen production [kg/year]                                                       |
| $\overline{G}$ :   | Global horizontal radiation [kWh/m <sup>2</sup> /day]                    | LF                        | Load following                                                                            |
| $\overline{G_d}$ : | Diffuse solar radiation [kWh/m <sup>2</sup> /day]                        | CC:                       | Cycle charging                                                                            |

paper proposes a techno-economic analysis of an innovative hybrid energy system (Grid/Fuel Cell/PV/Electrolyzer) for hydrogen and electricity generation in some communities in the context of renewable and sustainable energy transition.

### 1.1. Benin's energy systems

Benin is a country in West Africa, with about thirteen million inhabitants and an area of 114,763 km<sup>2</sup> [5]. The electricity access in Benin is approximately 42 % [6,7,8]. Concerning the renewable energy systems, some research works were carried out. Indeed, Odou et al. proposed a PV/diesel/battery (150 kW/62.5 kVA/637 kWh) hybrid power system as the optimized architecture for a sustainable rural electrification in Benin [9]. The authors demonstrated that the system was reliable, minimized battery requirements when compared to PV/battery system and significantly reduced CO<sub>2</sub> emissions. Mensah et al. evaluated the biogas potential of some energy systems in ten Beninese cities [10]. It was highlighted that the investigations were promising for seven municipalities. Akpahou et al. analyzed the techno-economic feasibility of a 10 MW grid-tied PV system in seven cities in Benin [11]. The authors showed that the photovoltaic system generated 13,222 MWh/year of electrical energy. A levelized cost of energy between US\$ 0.110/kWh and US\$ 0.125/kWh was also obtained.

### 1.2. Burkina Faso's energy systems

Burkina Faso is a country in West Africa, with about twenty-two million inhabitants and an area of 274,000 km<sup>2</sup>. The electrification rate is about 19 % [12,13,8]. Various works on renewable energy (RE) systems were studied and analyzed in the country to improve the electricity access. Yamegueu et al. studied the electricity production from PV/diesel hybrid systems without battery storage for off-grid locations in Burkina Faso [12]. The results indicated that the overall performance of the diesel generator was affected by the use of a PV system. Tatsidjoudoug et al. performed a techno-economic, social, and environmental analysis for hybrid systems including biofuel in Burkina Faso [14]. The investigators showed that the development of biofuel could be a benefit in the energy transition in the country. Ouedraogo et al. evaluated the

levelized cost of energy (LCOE) of a PV/Diesel hybrid system in the village of Pissila in Burkina Faso [15]. It was found that the system could provide access to electricity to the population. It was observed a decrease of 54 % in the LCOE when compared to the diesel generator. Baumert et al. carried out a study on the life cycle assessment of the Jatropha systems for biofuel generation [16]. It was concluded that the use of the Jatropha biofuel significantly minimized greenhouse gas production (between 68 and 89 %) and saved energy (between 65 and 90 %) when compared to diesel fuel. Sawadogo et al. produced the energy by developing cashew waste briquettes [17]. The analysis of the mechanical and physicochemical properties of the briquettes showed a net calorific value of 25.7 MJ/kg. Ouedraogo et Yamegueu performed a techno-economic evaluation of PV integration into national grids [18]. The investigators concluded that the architectures with PV integration were more attractive and affordable when compared with the diesel system. Abid et al. performed a techno-economic feasibility analysis of off-grid and on-grid solar PV systems with pumped hydro storage and batteries for the electrification of various cities [19]. It was suggested that the system made up of PV with pumped hydro storage remained the optimal system. Ouedraogo et al. analyzed a PV system performance for the electrification of a hospital [20]. It was demonstrated that the system could be utilized to ameliorate the population's resilience in high-security challenges areas.

### 1.3. Burundi's energy systems

Burundi is a country in East Africa with about thirteen million inhabitants and an area of 27,834 km<sup>2</sup>. The electricity access is approximately 10.2 % [21,22,8]. A sustainable planning of the energy sector in Burundi is still in progress [21]. Nevertheless, Manirambona et al. proposed a micro hydropower plant for the electrification of the city of Kigwena [23]. It was concluded that the study could be a replicable model for other areas facing similar challenges. Godfrey designed a solar PV kiosk in high-temperatures environments [24]. The investigator suggested the use of refrigeration cooling systems as appropriate for high-temperature environments such as the Ruhoro solar kiosk. Bime-nyimana et al. developed a techno-economic feasibility analysis for an affordable energy system in East African community countries [25].

#### 1.4. Cameroon's energy systems

Cameroon is a country in Central Africa, with about twenty-eight million inhabitants and an area of 475,442 km<sup>2</sup>. The electricity access is approximately 65.4 % [26,27–30,8]. Various works dealing with renewable energy systems were developed. Kenfack et al. highlighted a microhydro-PV hybrid system to generate electricity in a village in Cameroon. It was found that the system could be promoted in similar locations. Nfah and Ngundam explored the feasibility of pico-hydro-biogas and PV-biogas hybrid systems to supply energy in remote areas in Cameroon [31]. It was concluded that off-grid architectures based on local energy resources could provide adequate solutions to electrify rural areas in Cameroon. Muh and Tabet compared nine energy systems for rural applications in Southern Cameroons [32]. It was confirmed that the PV/Diesel/Hydro/Battery system was the most promising architecture for Southern Cameroons with a levelized cost of energy of US\$ 0.443/kWh. Yimen et al. developed an optimal design of a biomass-based hybrid energy system for the electrification of rural areas in Northern Cameroon [33]. The solutions suggested that the optimal architecture was a 98.1 kW PV, a 30 kW biogas generator, and 200 batteries, with a levelized cost of energy of US\$ 0.347/kWh. Tonsie et al. explored the techno-economic design of a Photovoltaic/Diesel/Batteries for electrification purposes in the city of Maroua [34]. The results revealed that the optimal configuration consisted of a 200 W PV and batteries of 200 Ah. Tiam et al. proposed an off-grid PV/FC/Biogas/Electrolyzer/Battery system for energy and hydrogen generation in the city of Maroua in Cameroon [35]. It was justified that the association of an electrolyzer, a fuel cell, and a hydrogen tank could be combined to biogas generator and PV panels for energy production. A techno-economic analysis of a PV/Battery/wind system for off-grid electrification was constructed [36]. The authors evaluated the impact of solar trackers in the performance of the system and found that the dual-axis tracker improved the electricity production. Techno-economic studies of a PV/wind system with various energy storage technologies for the electrification of three different cities in Cameroon were examined [37]. Ngouleu et al. performed a comparative study of meta-heuristics optimization techniques to find an optimal design of PV/wind/FC and PV/Battery/wind hybrid systems [38]. It was shown that the PV/Battery/wind hybrid system was the most reliable and affordable architecture to satisfy the electricity load profiles in the city of Kousseri in Cameroon.

#### 1.5. Chad's energy systems

Chad is a country in Central Africa, with about eighteen million inhabitants and an area of 1284 million km<sup>2</sup>. The electricity access is approximately 11.3 % [39,8]. Different energy systems were developed in Chad. Jahangiri et al. proposed a techno-economic and environmental design of a PV/wind hybrid system for electricity and hydrogen generation [39]. It was obtained a levelized cost of energy ranging between 0.507 and US\$ 0.573/kWh. Soulouknga et al. performed an economic study of the generation of electricity with various wind turbines [40]. It was suggested the utilization of a Vestas 2 MW/80 wind turbine for power production in Chad. Bongole et al. examined the possibility of generating power from an improved geothermal system [41]. The authors suggested that implementing the system could produce significant benefits in terms of technical, environmental, and economic analysis. Hassane et al. conducted a techno-economic study of a mini-grid PV system in Chad [42]. It was concluded that the cost of energy was more affordable when compared with the cost of energy proposed by the national electricity company. Soulouknga et al. assessed the performance of five wind generators in different locations in Chad. The results indicated that the Enercom E–82 wind turbine could be suitable for the sites of study [43]. A PV/Diesel/wind/battery hybrid system was proposed for the electrification of isolated locations in Chad [44]. The levelized cost of energy was between 0.367 and US\$ 0.529/kWh.

#### 1.6. Congo Republic's energy systems

Congo Republic is a country in Central Africa, with approximately 5.836 million inhabitants and an area of 342,000 km<sup>2</sup>. The electricity access is approximately 49.7 % [8]. No research work concerning renewable energy systems was developed in the country.

#### 1.7. Ivory Coast's energy systems

Ivory Coast is a country in West Africa, with about twenty-eight million inhabitants and an area of 322,462 km<sup>2</sup>. The electricity access is approximately 71.1 % [45,8]. The only work performed was the one developed by Pradas and Soucase, who examined the economic analysis of an on-grid 10 kW PV for electricity generation in Ivory Coast [45].

#### 1.8. Gabon's energy systems

Gabon is a country in Central Africa, with about 2.341 million inhabitants and an area of 267,668 km<sup>2</sup>. The electricity access is approximately 91.8 % [46,8]. No scientific publication about energy systems was found in the literature.

#### 1.9. Guinea's energy systems

Guinea is a country located in West Africa. It has about fourteen million inhabitants and an area of about 245,857 km<sup>2</sup>. The electrification rate is about 46.8 % [47,8]. A techno-economic evaluation of offshore wind energy potential at some locations in the gulf of Guinea was determined [47]. It was found that the levelized cost of energy ranged between 52.29 and US\$ 137.12/MWh.

#### 1.10. Guinea-bissau's energy systems

Guinea-Bissau is a country in West Africa, with about 2.061 million inhabitants, an area of 36,125 km<sup>2</sup> [48] and an electrification rate of about 33.3 % [49]. The only work performed was the one developed by Ianda and Padula who designed a very first institutional and organizational framework for biodiesel production in Guinea-Bissau and discovered that it could reduce poverty in the country [49].

#### 1.11. Equatorial Guinea's energy systems

Equatorial Guinea is a country in Central Africa, with about 1634 million inhabitants and an area of 28,051 km<sup>2</sup>. The electricity access is approximately 66.8 %. No scientific paper about energy systems was found in the literature.

#### 1.12. Madagascar's energy systems

Madagascar is an island located in the southeast coast of Africa. It has about twenty-nine million inhabitants, an area of 586,884 km<sup>2</sup>, and an electricity access of 35.8 % [50,8]. Garcia and Raji developed a techno-economic viability of a renewable energy system to provide electricity access to the population of Madagascar by using renewable resources [50]. It was proposed an architecture consisting of PV, wind turbines, and hydro components. A biogas technology was constructed and proposed for livestock farming in Madagascar [51].

#### 1.13. Mali's energy systems

Mali is a country in West Africa with approximately twenty-two million inhabitants, an area of 1,24 million km<sup>2</sup>, and an electricity access of 53.4 % [52,8]. Very little work was carried out in the field of energy systems. Touré et al. analyzed some hybrid energy systems (on-grid PV in particular) and demonstrated the capability of the system for rural electrification [52].

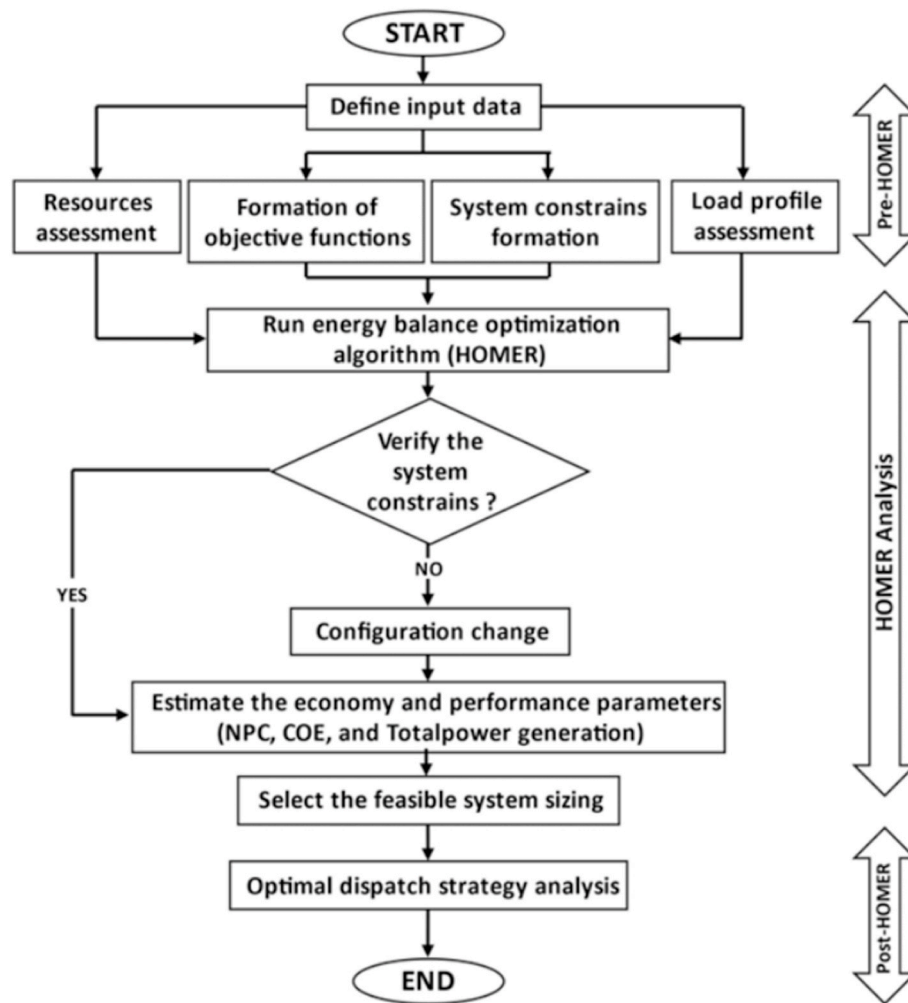


Fig. 1. Flowchart diagram of HOMER Software [87].

#### 1.14. Niger's energy systems

Niger is a country in West Africa with more than twenty-six million inhabitants, 1.267 million km<sup>2</sup>, and an electrification rate of 18.6 % [53, 8]. Very little work was conducted in the production of energy systems. Bandhari explored the green hydrogen production in Niger [53]. It was concluded that only 5 % of the land area in Niger could be sufficient to produce the required electricity from solar PV to generate hydrogen.

#### 1.15. Central African Republic's energy systems

Central African Republic is a country with 5.457 million inhabitants, an area of 623,000 km<sup>2</sup>, and an electricity access of 15.7 % [54,8]. Concerning the state-of-art of energy systems, Touafio et al. analyzed a wind generator project in the city of Bouar [54]. The simulations revealed that the cost of energy was about US\$ 0.081/kWh.

#### 1.16. Democratic Republic of Congo's energy systems

It is a country in Central Africa with almost ninety-seven million inhabitants, an area of 2.345 million km<sup>2</sup>, and an electricity access of 20.8 % [55,8]. A few works were tackled with the ultimate goal of generating electricity from renewable energy systems. Kusakana and Vermaak developed a hybrid renewable energy system for mobile telephony base stations in developing countries [55]. The system consisted of two 7.5 kW wind turbines, 10 kW PV, 7.5 kW inverter, and 82 batteries with a net present cost of 196,975 US\$ and a cost of energy of US\$

0.372/kWh. It was also designed a PV/wind charging station for small electric Tuk-tuk [56]. Hurtado et al. optimized a hybrid energy system (a combination of a PV, a biomass gasifier, and a battery bank) to supply energy in the « Institut Supérieur des Techniques Appliquées » [57]. It was indicated that the stability of the electrical energy produced could reach up to 98 % if the renewable energy sources were well-sized. Rice et al. evaluated the possibility of utilizing technical data to assess the efficiency of a hybrid system made up of a PV and a diesel power plant [58]. It was proposed a hybrid system for electricity production in the country.

#### 1.17. Rwanda's energy systems

It is a country in East Africa, with about 13.46 million inhabitants, and an area of 26,336 km<sup>2</sup>. It has an electrification rate of 48.7 % [59, 60–62,8]. Very few works were conducted in this country concerning energy systems. Niyonteze et al. examined various scenarios of hybrid systems for the electricity generation in some locations in Rwanda [59]. It was proposed the hydro/PV/battery system because it was cost-effective.

#### 1.18. Senegal's energy systems

Senegal is a country in West Africa with about seventeen million inhabitants, an area of 196,722 km<sup>2</sup>, and an electricity access of 68 % [63,64,8]. A few works were conducted in the country to improve this electrification rate. Alzola et al. proposed hybrid microgrids with high



**Table 1**  
Geographical coordinates of CAMES countries [89].

| Country                      | Latitude     | Longitude    | Sunniest city (site of study) |
|------------------------------|--------------|--------------|-------------------------------|
| Benin                        | 9.307690 N   | 2.315834 E   | Karimama                      |
| Burkina Faso                 | 12.238333 N  | −1.561593 W  | Sebba                         |
| Burundi                      | −3.3730560 S | 29.918886 E  | Kinyinya                      |
| Cameroon                     | 7.369722 N   | 12.354722 E  | Maroua                        |
| Central African Republic     | 6.611111 N   | 20.939444 E  | Birao                         |
| Chad                         | 15.454166 N  | 18.732207 E  | Amdjarass                     |
| Congo Republic               | −4.266130 S  | 15.283180 E  | Betou                         |
| Democratic Republic of Congo | −4.038333 S  | 21.758664 E  | Lubumbashi                    |
| Gabon                        | −0.803689 S  | 11.609444 E  | Lekoni                        |
| Guinea                       | 9.945587 N   | −9.696645 W  | Labe                          |
| Guinea-Bissau                | 11.803749 N  | −15.180413 W | Bafata                        |
| Equatorial Guinea            | 1.650801 N   | 10.267895 E  | Bata                          |
| Ivory Coast                  | 7.539989 N   | −5.547080 W  | Odienné                       |
| Madagascar                   | −18.766947 S | 46.869107 E  | Belo                          |
| Mali                         | 17.570692 N  | −3.996166 W  | Bouhessa                      |
| Niger                        | 17.607789 N  | 8.081666 E   | Madama                        |
| Rwanda                       | −1.940278 S  | 29.873888 E  | Kibuye                        |
| Senegal                      | 14.497401 N  | −14.452362 W | Dakar                         |
| Togo                         | 8.619543 N   | 0.824782 E   | Dapaong                       |

contents of energy for rural electrification [63].

#### 1.19. Togo's energy systems

Togo is a country in West Africa with about 8.645 million inhabitants, an area of 56,785 km<sup>2</sup>, and an electricity access of 55.7 % [8]. No scientific paper about energy systems was found in the literature.

Various other papers dealing with hybrid energy systems applied to different other countries in the world were developed (PV/Wind/Biomass [65]; PV/Wind, PV/Biomass, Wind/Biomass, and PV/Wind/Biomass systems [66–68,69,70]; PV/Wind turbine/Fuel cell/Battery [71]; Grid/PV/Battery [72,73]; PV/Wind/Diesel/Battery [74,75,76]; Biomass/Fuel cell [77]; PV/Fuel cell [78,79]; PV with different battery technologies [80,81,73]; PV/Wind [82]; PV/Wind/Fuel cell/Biomass [83]; PV/Diesel [84,73,76]; PV with hydrogen storage technology [85]; PV/Diesel/Grid [73], PV/Wind/Battery [76], PV/Natural gas/Battery [76], PV/Diesel/Natural gas/Battery [76], Wind/Natural gas/Battery [76], Wind/Diesel/Natural gas/Battery [76], PV/Biogas [86]).

It appeared from the literature review that renewable energy systems are poorly studied and implemented in developing countries. Therefore, they need to be developed and incorporated in the energy mix of the countries of CAMES as it was recommended during the scientific days of CAMES held in 2021 in Senegal. The proposed recommendation should lead to the establishment of a pragmatic strategy for the planning, development, and popularization of renewable energy systems that are affordable, viable, and capable of improving access to electricity in an approach of a sustainable energy transition in the countries of CAMES. That is why the present work proposes for the very first time a techno-economic analysis of a Grid/Fuel Cell/PV/Electrolyzer hybrid system for hydrogen and electricity generation in some communities of the nineteen countries of the African and Malagasy Council for Higher Education (Benin, Burkina Faso, Burundi, Cameroon, Central African Republic, Congo Republic, Ivory Coast, Gabon, Guinea, Guinea-Bissau, Equatorial Guinea, Madagascar, Mali, Niger, Democratic Republic of Congo, Rwanda, Senegal, Chad, and Togo) in a strategy of sustainable energy transition. The analysis was performed by using HOMER Pro software. The novelty of the paper is threefold: (i) The Grid/Fuel Cell/PV/Electrolyzer hybrid system is modeled, simulated, and optimized in each of the aforementioned countries for the very first time; (ii) The

levelized costs of hydrogen and energy were evaluated and compared; (iii) PV, fuel cell, grid purchases, and grid sales power productions were determined and compared. The rest of the document is divided as follows: The methodology is presented in section 2. Section 3 shows the results and discussions while section 4 develops some conclusions and perspectives of the work.

## 2. Material and methods

In the present work, HOMER (Hybrid Optimization of Multiple Energy Resources) software was used for designing, simulating and optimizing the Grid/Fuel Cell/PV/Electrolyzer hybrid system. Different energy components are proposed in the platform (solar PV, wind generators, various energy storage technologies, generators etc.). The energy and components resources, technical and economical constraints, energy storage technologies and the various control strategies are some of the basic information required in order to run a simulation. The flow diagram of HOMER is depicted in Fig. 1 [87].

### 2.1. Locations of CAMES countries

This study deals with nineteen countries, namely: Benin, Burkina Faso, Burundi, Cameroon, Central African Republic, Congo Republic, Ivory Coast, Gabon, Guinea, Guinea-Bissau, Equatorial Guinea, Madagascar, Mali, Niger, Democratic Republic of Congo, Rwanda, Senegal, Chad, and Togo [88]. The geographical coordinates of the aforementioned countries are given in Table 1 [89]. Fig. 2 presents the locations in the Africa map [90].

### 2.2. Daily load profile approximation

In the present work, a community of low consumers (consumers depending principally on farming) in each country was considered [35]. Indeed, we are mainly interested in the present paper in the populations of some villages with low access to electricity. The main use of electricity in each community is the lighting. Fig. 3 depicts the estimation of the daily load profile in each community. Table 2 gives some details concerning the appliances and the energies consumed in each community.

### 2.3. Solar radiation resources in the CAMES countries

The monthly average solar radiation resources for each country of the African and Malagasy Council for Higher Education were obtained from the National Aeronautics and Space Administration prediction of worldwide energy resources database [91]. Fig. 4 presents the monthly average solar radiation resources for each country. It was observed a maximum average solar radiation (about 7.5 kWh/m<sup>2</sup>/day) during the month of May in the city of Bouhessa in Mali and a minimum value (about 4.0 kWh/m<sup>2</sup>/day) during the month of December in the city of Madama in Niger.

Fig. 5 shows the solar resource map of Africa [92]. It was concluded from Figs. 4 and 5 that the solar radiation resources of the African and Malagasy Council for Higher Education were sufficient for the development of solar energy projects. Therefore, the present work explored the techno-economic analysis of a Grid/Fuel Cell/PV/Electrolyzer hybrid system for the generation of electricity and hydrogen in the aforementioned countries by using HOMER software.

### 2.4. HOMER procedure modeling

The architecture considered in this work was made up of photovoltaic panels, a converter, an electrolyzer, and a fuel cell, all connected to the electrical grid (see Fig. 6). A hydrogen tank was also added to the model for hydrogen storage. An external demand of hydrogen (0.25 kg/h) was used in this research as the hydrogen load for more applications purposes.



Fig. 2. Locations of CAMES countries in the Africa map [90].

#### 2.4.1. PV modeling procedure

The procedure for modeling photovoltaic panels in the software followed the Duffie and Beckman approach [93].

- Step 1: The solar declination was evaluated as follows:

$$\delta = 23.45^\circ \sin\left(360^\circ \left(\frac{284 + n}{365}\right)\right) \quad (1)$$

Where  $n$  is the day of the year (a number between 1 and 365);

- Step 2: The hour angle ( $\omega$ ) was computed from the equation of time (2) and the solar time (3):

$$t_s = t_c + \frac{\lambda}{15^\circ/\text{hr}} - Z_c + E \quad (3)$$

$$\omega = (t_s - 12\text{hr}) \bullet 15^\circ / \text{hr} \quad (4)$$

where  $t_c$  the civil time in hours corresponding to the midpoint of the time step;  $\lambda$  the longitude in degrees; and  $Z_c$  the time zone in hours of Greenwich Mean Time;

$$E = 3.82 \left( \begin{aligned} &0.000075 + 0.001868 \bullet \cos\left(360^\circ \left(\frac{n-1}{365}\right)\right) - 0.032077 \bullet \sin\left(360^\circ \left(\frac{n-1}{365}\right)\right) \\ &- 0.014615 \bullet \cos\left(720^\circ \left(\frac{n-1}{365}\right)\right) - 0.04089 \bullet \sin\left(720^\circ \left(\frac{n-1}{365}\right)\right) \end{aligned} \right) \quad (2)$$

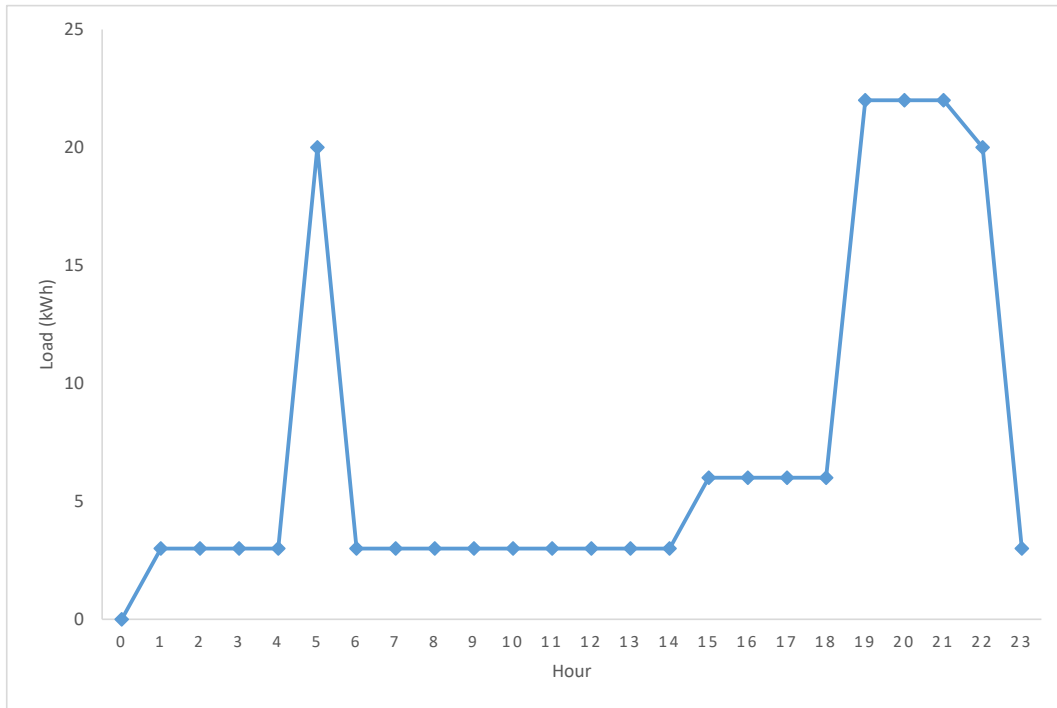


Fig. 3. Approximation of the daily load profile for a community of low consumers in each CAMES country [35].

**Table 2**

An overview of the daily load profile for a community of low consumers in each CAMES country [35].

| Equipment      | Power (W) | Quantity | Number of hours per day | Total power (W) | Energy (Wh)/day |
|----------------|-----------|----------|-------------------------|-----------------|-----------------|
| Phone charger  | 5         | 3        | 8.33                    | 15              | 125             |
| Radio          | 15        | 1        | 8                       | 15              | 120             |
| Security bulb  | 15        | 1        | 11                      | 15              | 165             |
| Lighting bulbs | 15        | 5        | 5                       | 75              | 375             |
| Random load    | 15        | 1        | 5                       | 15              | 75              |
| <b>Total</b>   |           |          |                         | <b>135</b>      | <b>860</b>      |

- Step 3: The angle of incidence  $\theta$  (angle between the radiation of sun's beam and the normal to the surface) was evaluated for a surface with any orientation:

$$\cos \theta = \sin \delta \sin \varphi \cos \beta - \sin \delta \cos \varphi \sin \beta \cos \gamma + \cos \delta \cos \varphi \cos \beta \cos \omega + \cos \delta \sin \varphi \sin \beta \cos \gamma \cos \omega + \cos \delta \sin \beta \sin \gamma \sin \omega \quad (5)$$

where  $\theta$ ,  $\beta$ ,  $\gamma$ ,  $\varphi$ ,  $\delta$  and  $\omega$  are the angle of incidence, the slope of the surface of the PV array, the azimuth of the surface, the latitude, the solar declination, and the hour angle in degrees.

- Step 4: The Zenith angle  $\theta_z$  was computed through the following equation:

$$\cos \theta_z = \cos \varphi \cos \delta \cos \omega + \sin \varphi \sin \delta \quad (6)$$

- Step 5: The extraterrestrial normal irradiance  $G_{on}$  (amount of solar radiation striking a surface normal to the sun's rays at the top of the earth's atmosphere) was now calculated:

$$G_{on} = 1.367 \left( 1 + 0.033 \cdot \cos \left( \frac{360n}{365} \right) \right) \quad (7)$$

- Step 6: The extraterrestrial horizontal irradiance  $G_0$  (amount of solar radiation striking a horizontal surface at the top of the atmosphere) was considered:

$$G_0 = G_{on} \cos \theta_z \quad (8)$$

- Step 7: The average extraterrestrial horizontal irradiance  $\overline{G_0}$  was evaluated:

$$\overline{G_0} = \frac{12}{\pi} G_{on} \left[ \cos \varphi \cos \delta (\sin \omega_2 - \sin \omega_1) + \frac{\pi(\omega_2 - \omega_1)}{180^\circ} \sin \varphi \sin \delta \right] \quad (9)$$

where  $\omega_1$  and  $\omega_2$  represented the hour angle at the beginning and at the end of the time step respectively (in degrees);

- Step 8: The clearness index  $k_T$  was found:

$$k_T = \frac{\overline{G}}{\overline{G_0}} \quad (10)$$

With  $\overline{G}$  the global horizontal radiation on the earth's surface averaged over the time step.

- Step 9: The diffuse radiation (solar radiation whose direction has been changed by the earth's atmosphere) was defined:

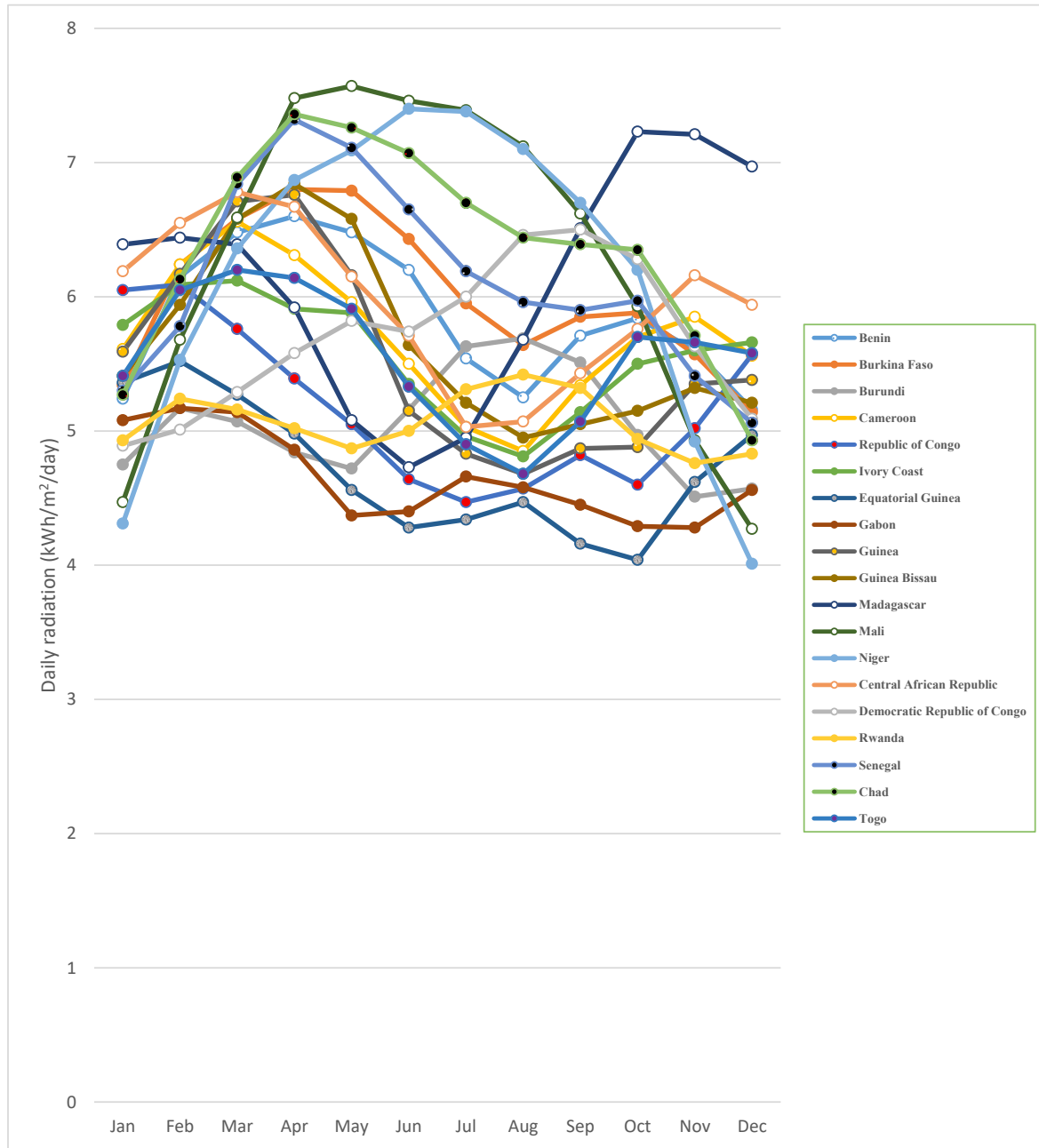


Fig. 4. Monthly average solar radiation in CAMES countries.

$$\frac{\overline{G}_d}{\overline{G}} = \begin{cases} 1.0 - 0.09 \bullet k_T & \text{for } k_T \leq 0.22 \\ 0.9511 - 0.1604 \bullet k_T + 4.388 \bullet k_T^2 - 16.638 \bullet k_T^3 + 12.336 \bullet k_T^4 & \text{for } 0.22 < k_T \leq 0.80 \\ 0.165 & \text{for } k_T > 0.80 \end{cases} \quad (11)$$

- Step 10: The ratio ( $R_b$ ) of beam radiation on the tilted surface to beam radiation on the horizontal surface, the anisotropy index  $A_i$  (a measure of the atmospheric transmittance of beam radiation) and a factor  $f$  used to account for horizon brightening were computed:

$$R_b = \frac{\cos \theta}{\cos \theta_z}, A_i = \frac{\overline{G}_b}{\overline{G}_0}, f = \sqrt{\frac{\overline{G}_b}{\overline{G}}} \quad (12)$$



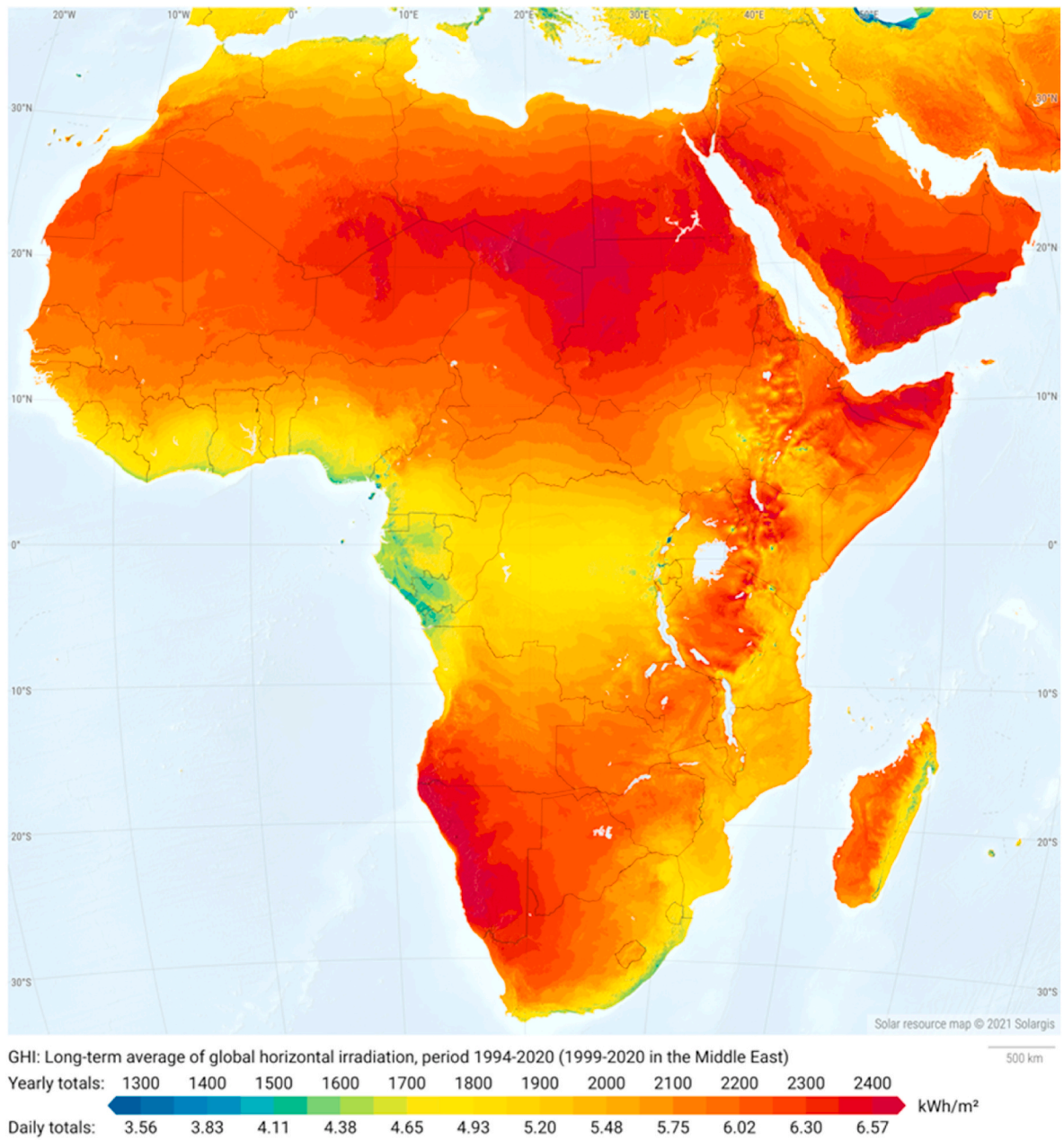


Fig. 5. Solar resource map of Africa [92].

- Step 10: Finally, the global radiation incident on the PV array to calculate the cell temperature and the power output of the PV array was used:

$$\begin{aligned} \overline{G_T} = & (\overline{G_b} + \overline{G_d} A_i) R_b + \overline{G_d} (1 - A_i) \left( \frac{1 + \cos \beta}{2} \right) \left[ 1 + f \sin^3 \left( \frac{\beta}{2} \right) \right] \\ & + \overline{G_{\rho_g}} \left( \frac{1 - \cos \beta}{2} \right) \end{aligned} \quad (13)$$

With  $\rho_g$  the ground reflectance, called the albedo [%].

The technical specifications and costs of the PV cells used in the work were defined in Table 3. The costs associated to the PV took into account the purchase outside the country, the customs, mounting hardware, wiring, installation, and maintenance.

#### 2.4.2. Electrolyzer

It is an equipment powered by electricity and based on the water electrolysis process to decompose water into oxygen and hydrogen gases

(see equation (14)).



The properties and costs of the electrolyzer introduced in HOMER were defined in Table 4.

#### 2.4.3. Fuel cell

A fuel cell is a device used for the conversion of hydrogen fuel into electrical energy. A generator component powered by the stored hydrogen was used and modeled in HOMER software. The properties and costs of the fuel cell used in HOMER were defined in Table 5.

#### 2.4.4. Converter

The present architecture modeled in HOMER software contained both AC and DC components. Therefore, a converter was required in the design. The properties and costs of the converter implemented in HOMER were defined in Table 6.

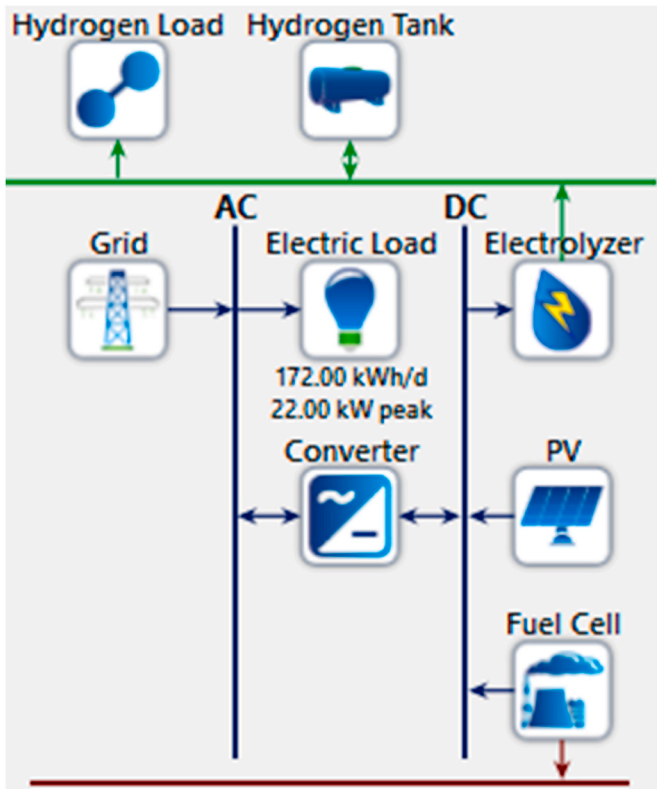


Fig. 6. Architecture implemented in HOMER software.

**Table 3**  
Technical specifications and costs of the PV cells [35].

|                                  |       |
|----------------------------------|-------|
| Maximum power (W)                | 1000  |
| Nominal voltage (V)              | 12    |
| Maximum power voltage (V)        | 17.5  |
| Maximum power current (A)        | 2.9   |
| Open-circuit voltage (V)         | 21.8  |
| Short-circuit current (A)        | 3.17  |
| Surface (m <sup>2</sup> )        | 0.451 |
| Capital cost (US\$)              | 724   |
| Replacement (US\$)               | 724   |
| Operation and maintenance (US\$) | 0.    |
| Lifetime (years)                 | 25    |
| Derating factor (%)              | 80    |

**Table 4**  
Properties and costs of the electrolyzer [35].

| Capacity (kW) | Capital (US\$) | Replacement (US\$) | Operation and maintenance (US\$/year) | Lifetime (years) | Efficiency (%) |
|---------------|----------------|--------------------|---------------------------------------|------------------|----------------|
| 1             | 1850           | 1573               | 30                                    | 15               | 74             |

**Table 5**  
Properties and costs of the fuel cell [35].

|                                             |         |
|---------------------------------------------|---------|
| Capacity (kW)                               | 1       |
| Capital (US\$)                              | 370     |
| Replacement (US\$)                          | 370     |
| Operation and maintenance (US\$)            | 0       |
| Lifetime (hours)                            | 50,000  |
| DC power consumption (kWh/Nm <sup>3</sup> ) | 3.9–4.4 |
| Dionized water consumption (l/h)            | 900     |
| Hydrogen production (Nm <sup>3</sup> /h)    | 0–1000  |

**Table 6**  
Properties and costs of the converter [35].

| Capacity (kW) | Capital (US\$) | Replacement (US\$) | Operation and maintenance (US\$) | Lifetime (years) | Efficiency (%) |
|---------------|----------------|--------------------|----------------------------------|------------------|----------------|
| 1             | 184            | 184                | 0                                | 20               | 95             |

**Table 7**  
Properties and costs of the Hydrogen tank [35].

| Size (kg) | Capital (US\$) | Replacement (US\$) | Operation and maintenance (US\$) | Lifetime (years) |
|-----------|----------------|--------------------|----------------------------------|------------------|
| 1         | 734            | 734                | 0                                | 25               |

**Table 8**  
Average grid power price used in HOMER for each CAMES country [94].

| Country                      | Average grid power price (US\$/kWh) |
|------------------------------|-------------------------------------|
| Benin                        | 0.260                               |
| Burkina Faso                 | 0.206                               |
| Burundi                      | 0.210                               |
| Cameroon                     | 0.083                               |
| Central African Republic     | 0.200                               |
| Chad                         | 0.330                               |
| Congo Republic               | 0.170                               |
| Democratic Republic of Congo | 0.067                               |
| Gabon                        | 0.204                               |
| Guinea                       | 0.350                               |
| Guinea-Bissau                | 0.230                               |
| Equatorial Guinea            | 0.210                               |
| Ivory Coast                  | 0.121                               |
| Madagascar                   | 0.130                               |
| Mali                         | 0.219                               |
| Niger                        | 0.190                               |
| Rwanda                       | 0.216                               |
| Senegal                      | 0.172                               |
| Togo                         | 0.193                               |

**Table 9**  
Sensitivity variables implemented in the present work [35].

| Sensitivity parameter     | Value                                    |
|---------------------------|------------------------------------------|
| Nominal discount rate (%) | {8.00; 9.20; 10.40; 11.60; 12.80; 14.00} |
| Project lifetime (years)  | {15, 17, 19, 21, 23, 25}                 |

#### 2.4.5. Hydrogen tank

This device was used to store the Hydrogen produced by the Electrolyzer. The properties and costs of the Hydrogen tank implemented in HOMER were defined in Table 7.

#### 2.4.6. Grid

The Grid component was also inserted in the architecture. The grid power price (in US\$/kWh) in each CAMES country was considered [94]. Table 8 presents the average grid power price used in HOMER for each country.

### 2.5. HOMER economic outputs

The optimization process in HOMER was mainly based on some economic outputs: Net Present Cost (NPC), levelized cost of Hydrogen (LCOH) and levelized cost of energy (LCOE). The NPC reads [35]:

$$NPC = \frac{C_{annualized}}{CRF(i, R_{proj})} \quad (15)$$

with  $CRF$ ,  $C_{annualized}$ ,  $i$  and  $R_{proj}$  representing the capital recovery factor,

**Table 10**  
Optimization results for Benin.

| Item              | Description    | Unit     | Value   |
|-------------------|----------------|----------|---------|
| Cost              | LCOE           | US\$/kWh | 0.275   |
|                   | NPC            | US\$     | 199,267 |
|                   | LCOH           | US\$/kg  | 2.00    |
| Power production  | PV             | kWh/year | 46,748  |
|                   | Fuel cell      | kWh/year | 43,751  |
|                   | Grid purchases | kWh/year | 30,194  |
| Power consumption | Grid sales     | kWh/year | 23,249  |
| Penetration       | PV             | %        | 74.50   |

**Table 11**  
Optimization results for Burkina Faso.

| Item              | Description    | Unit     | Value   |
|-------------------|----------------|----------|---------|
| Cost              | LCOE           | US\$/kWh | 0.258   |
|                   | NPC            | US\$     | 192,581 |
|                   | LCOH           | US\$/kg  | 1.94    |
| Power production  | PV             | kWh/year | 48,712  |
|                   | Fuel cell      | kWh/year | 43,754  |
|                   | Grid purchases | kWh/year | 30,603  |
| Power consumption | Grid sales     | kWh/year | 25,526  |
| Penetration       | PV             | %        | 77.60   |

**Table 12**  
Optimization results for Burundi.

| Item              | Description    | Unit     | Value   |
|-------------------|----------------|----------|---------|
| Cost              | LCOE           | US\$/kWh | 0.320   |
|                   | NPC            | US\$     | 210,529 |
|                   | LCOH           | US\$/kg  | 2.12    |
| Power production  | PV             | kWh/year | 37,899  |
|                   | Fuel cell      | kWh/year | 43,713  |
|                   | Grid purchases | kWh/year | 30,632  |
| Power consumption | Grid sales     | kWh/year | 15,250  |
| Penetration       | PV             | %        | 60.40   |

**Table 13**  
Optimization results for Cameroon.

| Item              | Description    | Unit     | Value   |
|-------------------|----------------|----------|---------|
| Cost              | LCOE           | US\$/kWh | 0.263   |
|                   | NPC            | US\$     | 188,387 |
|                   | LCOH           | US\$/kg  | 1.89    |
| Power production  | PV             | kWh/year | 45,237  |
|                   | Fuel cell      | kWh/year | 43,637  |
|                   | Grid purchases | kWh/year | 30,559  |
| Power consumption | Grid sales     | kWh/year | 22,070  |
| Penetration       | PV             | %        | 72.10   |

**Table 14**  
Optimization results for Central African Republic.

| Item              | Description    | Unit     | Value   |
|-------------------|----------------|----------|---------|
| Cost              | LCOE           | US\$/kWh | 0.263   |
|                   | NPC            | US\$     | 193,836 |
|                   | LCOH           | US\$/kg  | 1.94    |
| Power production  | PV             | kWh/year | 47,802  |
|                   | Fuel cell      | kWh/year | 43,752  |
|                   | Grid purchases | kWh/year | 30,419  |
| Power consumption | Grid sales     | kWh/year | 24,477  |
| Penetration       | PV             | %        | 76.10   |

**Table 15**  
Optimization results for Chad.

| Item              | Description    | Unit     | Value   |
|-------------------|----------------|----------|---------|
| Cost              | LCOE           | US\$/kWh | 0.247   |
|                   | NPC            | US\$     | 192,681 |
|                   | LCOH           | US\$/kg  | 1.94    |
| Power production  | PV             | kWh/year | 52,961  |
|                   | Fuel cell      | kWh/year | 43,784  |
|                   | Grid purchases | kWh/year | 30,314  |
| Power consumption | Grid sales     | kWh/year | 29,303  |
| Penetration       | PV             | %        | 84.40   |

**Table 16**  
Optimization results for Congo Republic.

| Item              | Description    | Unit     | Value   |
|-------------------|----------------|----------|---------|
| Cost              | LCOE           | US\$/kWh | 0.305   |
|                   | NPC            | US\$     | 203,845 |
|                   | LCOH           | US\$/kg  | 2.03    |
| Power production  | PV             | kWh/year | 38,988  |
|                   | Fuel cell      | kWh/year | 43,678  |
|                   | Grid purchases | kWh/year | 30,700  |
| Power consumption | Grid sales     | kWh/year | 16,314  |
| Penetration       | PV             | %        | 62.10   |

**Table 17**  
Optimization results for Democratic Republic of Congo.

| Item              | Description    | Unit     | Value   |
|-------------------|----------------|----------|---------|
| Cost              | LCOE           | US\$/kWh | 0.257   |
|                   | NPC            | US\$     | 184,844 |
|                   | LCOH           | US\$/kg  | 1.86    |
| Power production  | PV             | kWh/year | 45,942  |
|                   | Fuel cell      | kWh/year | 43,526  |
|                   | Grid purchases | kWh/year | 30,493  |
| Power consumption | Grid sales     | kWh/year | 22,571  |
| Penetration       | PV             | %        | 73.20   |

**Table 18**  
Optimization results for Gabon.

| Item              | Description    | Unit     | Value   |
|-------------------|----------------|----------|---------|
| Cost              | LCOE           | US\$/kWh | 0.344   |
|                   | NPC            | US\$     | 216,115 |
|                   | LCOH           | US\$/kg  | 2.17    |
| Power production  | PV             | kWh/year | 34,044  |
|                   | Fuel cell      | kWh/year | 43,708  |
|                   | Grid purchases | kWh/year | 30,654  |
| Power consumption | Grid sales     | kWh/year | 11,604  |
| Penetration       | PV             | %        | 54.20   |

**Table 19**  
Optimization results for Guinea.

| Item              | Description    | Unit     | Value   |
|-------------------|----------------|----------|---------|
| Cost              | LCOE           | US\$/kWh | 0.303   |
|                   | NPC            | US\$     | 212,769 |
|                   | LCOH           | US\$/kg  | 2.14    |
| Power production  | PV             | kWh/year | 43,788  |
|                   | Fuel cell      | kWh/year | 43,776  |
|                   | Grid purchases | kWh/year | 30,193  |
| Power consumption | Grid sales     | kWh/year | 20,459  |
| Penetration       | PV             | %        | 69.70   |

**Table 20**  
Optimization results for Guinea-Bissau.

| Item              | Description    | Unit     | Value   |
|-------------------|----------------|----------|---------|
| Cost              | LCOE           | US\$/kWh | 0.282   |
|                   | NPC            | US\$     | 200,490 |
|                   | LCOH           | US\$/kg  | 2.02    |
| Power production  | PV             | kWh/year | 44,897  |
|                   | Fuel cell      | kWh/year | 43,749  |
|                   | Grid purchases | kWh/year | 30,137  |
| Power consumption | Grid sales     | kWh/year | 21,430  |
| Penetration       | PV             | %        | 71.50   |

**Table 21**  
Optimization results for Equatorial Guinea.

| Item              | Description    | Unit     | Value   |
|-------------------|----------------|----------|---------|
| Cost              | LCOE           | US\$/kWh | 0.342   |
|                   | NPC            | US\$     | 216,038 |
|                   | LCOH           | US\$/kg  | 2.17    |
| Power production  | PV             | kWh/year | 34,681  |
|                   | Fuel cell      | kWh/year | 43,695  |
|                   | Grid purchases | kWh/year | 30,554  |
| Power consumption | Grid sales     | kWh/year | 12,093  |
| Penetration       | PV             | %        | 55.20   |

**Table 22**  
Optimization results for Ivory Coast.

| Item              | Description    | Unit     | Value   |
|-------------------|----------------|----------|---------|
| Cost              | LCOE           | US\$/kWh | 0.276   |
|                   | NPC            | US\$     | 193,129 |
|                   | LCOH           | US\$/kg  | 1.94    |
| Power production  | PV             | kWh/year | 43,616  |
|                   | Fuel cell      | kWh/year | 43,642  |
|                   | Grid purchases | kWh/year | 30,315  |
| Power consumption | Grid sales     | kWh/year | 20,290  |
| Penetration       | PV             | %        | 69.50   |

**Table 23**  
Optimization results for Madagascar.

| Item              | Description    | Unit     | Value   |
|-------------------|----------------|----------|---------|
| Cost              | LCOE           | US\$/kWh | 0.238   |
|                   | NPC            | US\$     | 183,536 |
|                   | LCOH           | US\$/kg  | 1.84    |
| Power production  | PV             | kWh/year | 52,238  |
|                   | Fuel cell      | kWh/year | 43,655  |
|                   | Grid purchases | kWh/year | 30,393  |
| Power consumption | Grid sales     | kWh/year | 28,572  |
| Penetration       | PV             | %        | 83.20   |

**Table 24**  
Optimization results for Mali.

| Item              | Description    | Unit     | Value   |
|-------------------|----------------|----------|---------|
| Cost              | LCOE           | US\$/kWh | 0.240   |
|                   | NPC            | US\$     | 187,181 |
|                   | LCOH           | US\$/kg  | 1.88    |
| Power production  | PV             | kWh/year | 53,176  |
|                   | Fuel cell      | kWh/year | 43,748  |
|                   | Grid purchases | kWh/year | 30,448  |
| Power consumption | Grid sales     | kWh/year | 29,608  |
| Penetration       | PV             | %        | 84.70   |

the annualized cost, the annual real discount rate (%), and the project lifetime (years). The leveled cost of energy was given in the present work as follows [35]:

**Table 25**  
Optimization results for Niger.

| Item              | Description    | Unit     | Value   |
|-------------------|----------------|----------|---------|
| Cost              | LCOE           | US\$/kWh | 0.241   |
|                   | NPC            | US\$     | 186,782 |
|                   | LCOH           | US\$/kg  | 1.88    |
| Power production  | PV             | kWh/year | 52,450  |
|                   | Fuel cell      | kWh/year | 43,751  |
|                   | Grid purchases | kWh/year | 30,433  |
| Power consumption | Grid sales     | kWh/year | 28,907  |
| Penetration       | PV             | %        | 83.50   |

**Table 26**  
Optimization results for Rwanda.

| Item              | Description    | Unit     | Value   |
|-------------------|----------------|----------|---------|
| Cost              | LCOE           | US\$/kWh | 0.321   |
|                   | NPC            | US\$     | 211,310 |
|                   | LCOH           | US\$/kg  | 2.13    |
| Power production  | PV             | kWh/year | 37,905  |
|                   | Fuel cell      | kWh/year | 43,713  |
|                   | Grid purchases | kWh/year | 30,616  |
| Power consumption | Grid sales     | kWh/year | 15,236  |
| Penetration       | PV             | %        | 60.40   |

**Table 27**  
Optimization results for Senegal.

| Item              | Description    | Unit     | Value   |
|-------------------|----------------|----------|---------|
| Cost              | LCOE           | US\$/kWh | 0.252   |
|                   | NPC            | US\$     | 189,095 |
|                   | LCOH           | US\$/kg  | 1.90    |
| Power production  | PV             | kWh/year | 49,875  |
|                   | Fuel cell      | kWh/year | 43,738  |
|                   | Grid purchases | kWh/year | 29,976  |
| Power consumption | Grid sales     | kWh/year | 25,987  |
| Penetration       | PV             | %        | 79.40   |

**Table 28**  
Optimization results for Togo.

| Item              | Description    | Unit     | Value   |
|-------------------|----------------|----------|---------|
| Cost              | LCOE           | US\$/kWh | 0.283   |
|                   | NPC            | US\$     | 199,440 |
|                   | LCOH           | US\$/kg  | 2.01    |
| Power production  | PV             | kWh/year | 43,727  |
|                   | Fuel cell      | kWh/year | 43,753  |
|                   | Grid purchases | kWh/year | 30,601  |
| Power consumption | Grid sales     | kWh/year | 20,788  |
| Penetration       | PV             | %        | 69.70   |

$$LCOE = \frac{C_{\text{annualized, total}}}{E_{\text{served}}} \quad (16)$$

$$LCOH = \frac{C_{\text{annualized, total}} - \nu_{\text{elec}} (E_{\text{prim, AC}} + E_{\text{prim, DC}} + E_{\text{Def}} + E_{\text{grid, sales}})}{M_{\text{Hydrogen}}} \quad (17)$$

Where  $C_{\text{annualized, total}}$  and  $E_{\text{served}}$  are the total annualized cost of the system (US\$/year) and the total electrical load served (kWh/year).  $\nu_{\text{elec}}$  is the value of electricity,  $E_{\text{prim}}$  is the primary electrical load,  $E_{\text{Def}}$  is the deferrable load,  $E_{\text{grid, sales}}$  is the total energy sold to the grid, and  $M_{\text{Hydrogen}}$  is the total hydrogen production.

## 2.6. Sensitivity variables and dispatch strategy

Two sensitivity variables were considered in the procedure of simulating and optimizing the architecture in HOMER software [35].

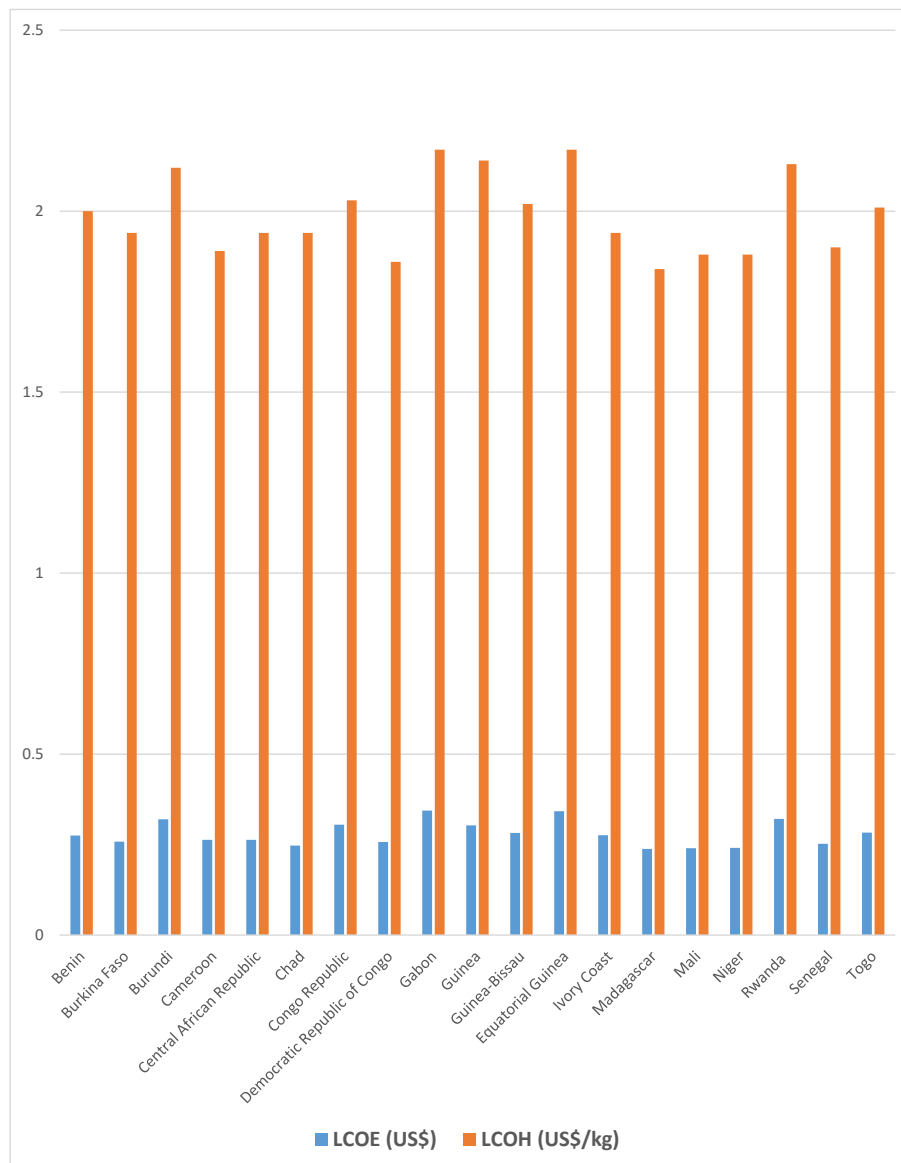


Fig. 7. Levelized cost of energy (LCOE) and levelized cost of Hydrogen (LCOH) for the hybrid system in the CAMES countries.

Table 9 lists the sensitivity variables. Two dispatch strategy which are defined as rules used for controlling a generator and the storage bank operation whenever there is insufficient renewable energy to supply the load were implemented: load following (LF) and cycle charging (CC) [35].

### 3. Results and discussions

A techno-economic analysis of a Grid/Fuel Cell/PV/Electrolyzer hybrid system for hydrogen and electricity generation in some communities of the nineteen countries of the African and Malagasy Council for Higher Education (Benin, Burkina Faso, Burundi, Cameroon, Central African Republic, Congo Republic, Ivory Coast, Gabon, Guinea, Guinea-Bissau, Equatorial Guinea, Madagascar, Mali, Niger, Democratic Republic of Congo, Rwanda, Senegal, Chad, and Togo) was performed. The optimized architecture obtained in each country was made up of a 20 kW PV, a 5 kW fuel cell, a 15 kW electrolyzer, a 175 kg Hydrogen tank, a 21.5 kW converter, and a 1000 kW grid. In addition, a CC dispatch strategy was found to be the best in all the countries. Tables 10–28 summarized and compared the main HOMER outputs for each CAMES country. The main HOMER outputs considered in the present study

consisted of general costs (levelized cost of energy, net present cost, and levelized cost of hydrogen); power productions (photovoltaic, fuel cell, and grid purchases); power consumption (grid sales) and the photovoltaic penetration. The grid purchases represented the maximum power that could be purchased from the grid while the grid sales were the maximum power that could be sold back to the grid. The photovoltaic penetration was defined as the ratio of PV electrical power output in a time step to the total electrical load served in this time step. The results presented in Tables 10–28 were summarized and compared in Figs. 7–10.

#### 3.1. Levelized cost of energy (LCOE)

It represents the average price of the unit (kWh) produced by an equipment. It was computed as the ratio of the total annualized price of the equipment to the total electrical load served. It takes into account capital costs, depreciation, operation, maintenance, management etc. It is used as a direct comparison of the costs of all kinds of electrical power generation. An average levelized cost of energy of US\$ 0.279/kWh was demonstrated. The exploitation of Tables 10–28 showed that the minimum and maximum LCOE were US\$ 0.238/kWh (in Madagascar) and



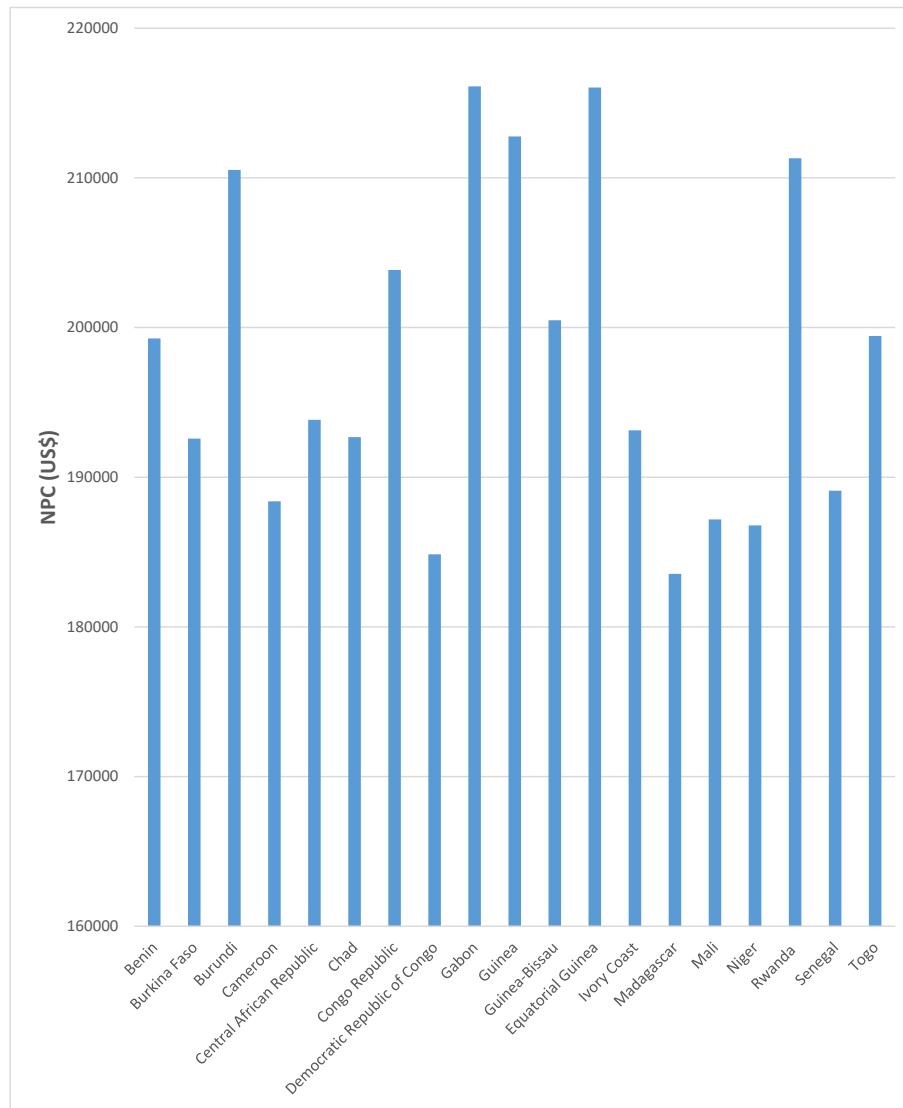


Fig. 8. Net present cost of the hybrid system in the CAMES countries.

US\$ 0.344/kWh (in Gabon) respectively (see Table 29 and Fig. 7). Indeed, Madagascar was followed by Mali, Niger, Chad, Senegal, Democratic Republic of Congo, Burkina-Faso, Cameroon, Central African Republic, Benin, Ivory Coast, Guinea-Bissau, Togo, Guinea, Congo Republic, Burundi, Rwanda, Equatorial Guinea, and Gabon. It was observed that the country with the lowest LCOE in Central Africa was Chad followed by the Democratic Republic of Congo, Cameroon, Central African Republic, Congo Republic, Equatorial Guinea, and Gabon. In West Africa, the country presenting the lowest LCOE was Mali followed by Niger, Senegal, Burkina Faso, Benin, Ivory Coast, Guinea-Bissau, Togo, and Guinea. In East Africa, Burundi was the country with the lowest LCOE, followed by Rwanda. The solutions confirmed that the levelized cost of energy was the highest in East Africa (about US\$ 0.321/kWh) followed by Central Africa (about US\$ 0.288/kWh), West Africa (about US\$ 0.268/kWh), and South East Africa (about US\$ 0.238/kWh). In addition, the implementation of the proposed hybrid system could be realistic in all the countries and more profitable in Madagascar particularly since the costs of energy were acceptable (considering the standard of living in these countries). In addition, it is important to highlight that higher solar radiation intensity corresponds to reduced costs, emphasizing the necessity of accurate solar radiation estimation for optimal system performance and cost efficiency. The LCOEs obtained in the present work are in the same range with those proposed in Ref. [35]

for the special case of Cameroon where a similar architecture was proposed.

### 3.2. Levelized cost of hydrogen (LCOH)

It is defined as the average price of Hydrogen produced by an equipment per unit mass. The quantity of Hydrogen stored inside a tank can be computed from the rate of incoming fuel or the rate of outgoing fuel. It also depends on the initial quantity of Hydrogen inside the tank. Hydrogen serves as a long-term energy storage, enabling the storage of energy for extended durations, potentially lasting weeks or even months. It offers the ability to store energy for scenarios involving seasonal variations or remote locations. However, it is important to consider that Hydrogen storage introduces added complexity and cost. An average levelized cost of Hydrogen of US\$ 1.99/kg was obtained. The minimum and maximum LCOH were US\$ 1.84/kg (in Madagascar) and US\$ 2.17/kg (in Gabon and Equatorial Guinea) respectively (see Table 30 and Fig. 7). Madagascar was followed by the Democratic Republic of Congo, Mali, Niger, Cameroon, Senegal, Burkina-Faso, Central African Republic, Chad, Ivory Coast, Benin, Togo, Guinea-Bissau, Congo Republic, Burundi, Rwanda, Guinea, Equatorial Guinea, and Gabon. It was depicted that the country with the lowest LCOH in Central Africa was the Democratic Republic of Congo followed by Cameroon, Central

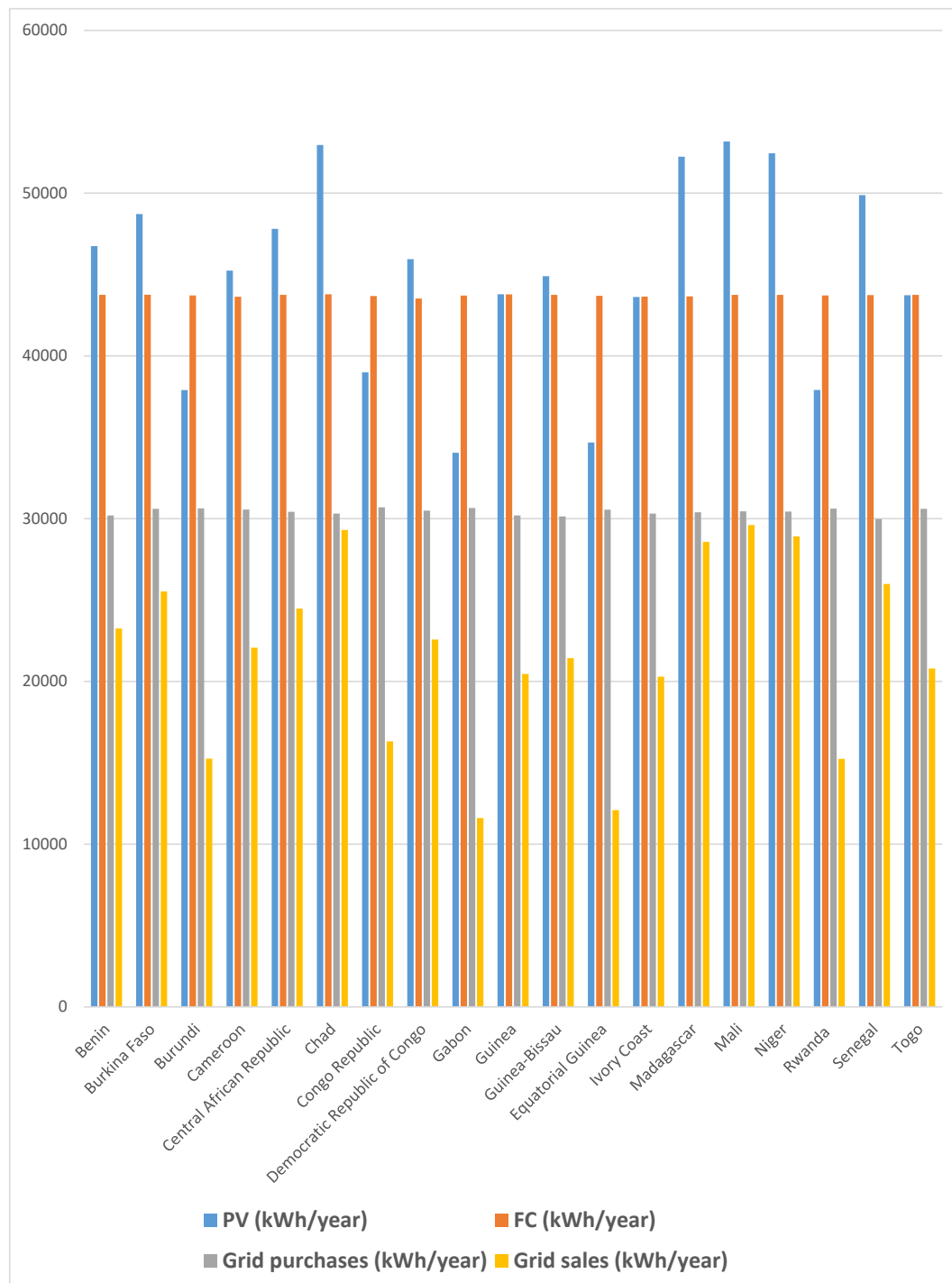


Fig. 9. Power production for each component of the hybrid system in the CAMES countries.

African Republic, Chad, Congo Republic, Equatorial Guinea, and Gabon. In West Africa, the country having the lowest LCOH was Mali followed by Niger, Senegal, Burkina Faso, Ivory Coast, Benin, Togo, Guinea-Bissau, and Guinea. Finally, in East Africa, Burundi was the country with the lowest LCOH, followed by Rwanda. The results suggested that the levelized cost of Hydrogen was the highest in East Africa (about US\$ 2.13/kg) followed by Central Africa (about US\$ 2.00/kg), West Africa (about US\$ 1.96/kg), and South East Africa (about US\$ 1.84/kg). The solutions were in good agreement with the LCOE since it was also revealed that the hybrid system could be promoted in these countries. The values of LCOH computed in the present study are in the same range

with those proposed in Ref. [35] for the special case of Cameroon where a similar architecture was proposed. These findings can inform the design and implementation of hydrogen-based energy systems for residential buildings, highlighting the importance of optimizing the use of solar panels and fuel cells to generate electricity, as well as the production of hydrogen using the electrolyzer, to achieve more sustainable and efficient energy consumption patterns.

### 3.3. Net present cost (NPC)

The NPC can be defined as the project costs' present value minus all

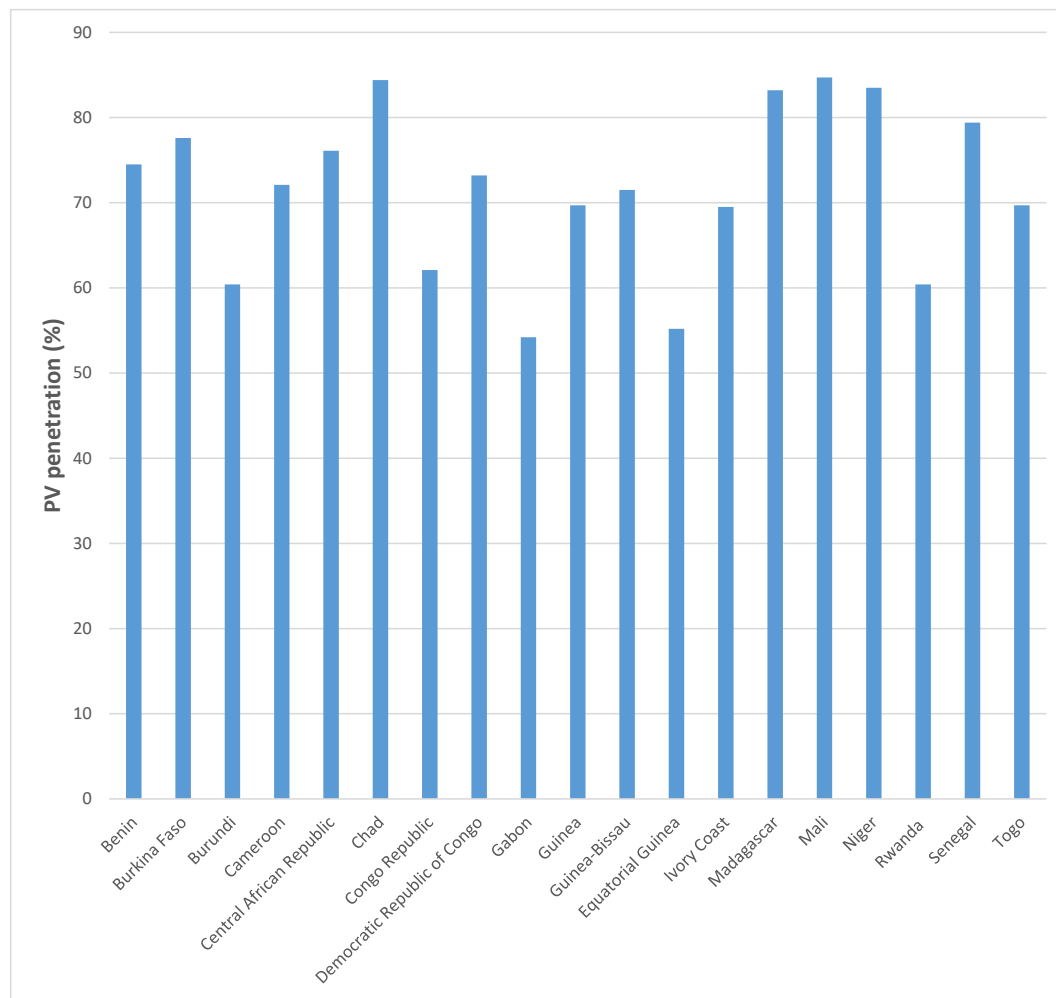


Fig. 10. PV penetration for the hybrid system in the CAMES countries.

**Table 29**  
Ranking of LCOE (US\$/kWh).

| Country                      | LCOE (US\$/kWh) | Rank |
|------------------------------|-----------------|------|
| Madagascar                   | 0.238           | 1    |
| Mali                         | 0.240           | 2    |
| Niger                        | 0.241           | 3    |
| Chad                         | 0.247           | 4    |
| Senegal                      | 0.252           | 5    |
| Democratic Republic of Congo | 0.257           | 6    |
| Burkina Faso                 | 0.258           | 7    |
| Cameroon                     | 0.263           | 8    |
| Central African Republic     | 0.263           | 9    |
| Benin                        | 0.275           | 10   |
| Ivory Coast                  | 0.276           | 11   |
| Guinea-Bissau                | 0.282           | 12   |
| Togo                         | 0.283           | 13   |
| Guinea                       | 0.303           | 14   |
| Congo Republic               | 0.305           | 15   |
| Burundi                      | 0.320           | 16   |
| Rwanda                       | 0.321           | 17   |
| Equatorial Guinea            | 0.342           | 18   |
| Gabon                        | 0.344           | 19   |

**Table 30**  
Ranking of LCOH.

| Country                      | LCOH (US\$/kg) | Rank |
|------------------------------|----------------|------|
| Madagascar                   | 1.84           | 1    |
| Democratic Republic of Congo | 1.86           | 2    |
| Mali                         | 1.88           | 3    |
| Niger                        | 1.88           | 4    |
| Cameroon                     | 1.89           | 5    |
| Senegal                      | 1.90           | 6    |
| Burkina Faso                 | 1.94           | 7    |
| Central African Republic     | 1.94           | 8    |
| Chad                         | 1.94           | 9    |
| Ivory Coast                  | 1.94           | 10   |
| Benin                        | 2.00           | 11   |
| Togo                         | 2.01           | 12   |
| Guinea-Bissau                | 2.02           | 13   |
| Congo Republic               | 2.03           | 14   |
| Burundi                      | 2.12           | 15   |
| Rwanda                       | 2.13           | 16   |
| Guinea                       | 2.14           | 17   |
| Gabon                        | 2.17           | 18   |
| Equatorial Guinea            | 2.17           | 19   |

profits over its expected lifetime. In the present situation, the costs of capital, replacement, operation, and maintenance are included in the project costs. The ranking of the computed net present costs showed that the minimum and maximum NPC were 183,536 US\$ (in Madagascar) and 216,115 US\$ (in Gabon) with an average value of 197,992.368 US\$

(see Table 31 and Fig. 8). Madagascar was followed by the Democratic Republic of Congo, Niger, Mali, Cameroon, Senegal, Burkina-Faso, Chad, Ivory Coast, Central African Republic, Benin, Togo, Guinea-Bissau, Congo Republic, Burundi, Rwanda, Guinea, Equatorial Guinea, and Gabon. It was demonstrated that the country with the lowest NPC in

**Table 31**  
Ranking of NPC.

| Country                      | NPC (US\$) | Rank |
|------------------------------|------------|------|
| Madagascar                   | 183,536    | 1    |
| Democratic Republic of Congo | 184,844    | 2    |
| Niger                        | 186,782    | 3    |
| Mali                         | 187,181    | 4    |
| Cameroon                     | 188,387    | 5    |
| Senegal                      | 189,095    | 6    |
| Burkina Faso                 | 192,581    | 7    |
| Chad                         | 192,681    | 8    |
| Ivory Coast                  | 193,129    | 9    |
| Central African Republic     | 193,836    | 10   |
| Benin                        | 199,267    | 11   |
| Togo                         | 199,440    | 12   |
| Guinea-Bissau                | 200,490    | 13   |
| Congo Republic               | 203,845    | 14   |
| Burundi                      | 210,529    | 15   |
| Rwanda                       | 211,310    | 16   |
| Guinea                       | 212,769    | 17   |
| Equatorial Guinea            | 216,038    | 18   |
| Gabon                        | 216,115    | 19   |

Central Africa was the Democratic Republic of Congo followed by Cameroon, Chad, Central African Republic, Congo Republic, Equatorial Guinea, and Gabon. In West Africa, the country with the lowest NPC was Niger followed by Mali, Senegal, Burkina Faso, Ivory Coast, Benin, Togo, Guinea-Bissau, and Guinea. Finally, in East Africa, Burundi was the country demonstrating the lowest NPC, followed by Rwanda. The solutions revealed that the NPC was the highest in East Africa (about 210,920 US\$) followed by Central Africa (about 199,392 US\$), West Africa (about 195,637 US\$), and South East Africa (about 183,536 US\$). One approach to reducing the system's cost involves improving the efficiency of hydrogen storage methods through research and technological advancements, consequently reducing infrastructure costs associated with storage.

### 3.4. Power production

The power productions for PV, fuel cell, grid sales and grid purchases were also determined and compared in Tables 10–28 and Fig. 9. It appeared from the comparison that the power production of PV, fuel cell, grid sales and grid purchases were between 34,044 (in Gabon) and 53,176 kWh/year (in Mali) for PV; between 43,526 (in Democratic Republic of Congo) and 43,784 kWh/year (in Chad) for fuel cell; between 29,976 (in Senegal) and 30,700 kWh/year (in Congo Republic) for grid purchases with average values of 44,983.4737 kWh/year, 43,711.7368 kWh/year and 30,433.3684 kWh/year respectively. The results suggested that the PV power production was the highest power generated by the hybrid system when compared with fuel cell, grid sales, and grid purchases power productions. Indeed, the highest PV electrical output was recorded in Mali, which has the highest average solar radiation of around 7.6 kWh/m<sup>2</sup>/day. This indicates that the solar panels were able to capture and convert more solar energy into electricity. Conversely, in Gabon, which has the lowest average solar radiation of about 4.2 kWh/m<sup>2</sup>/day, the PV electrical production decreased to the lowest amount of the year. This suggests that in Gabon, renewable energy systems are not able to produce as much electricity, highlighting the need for complementary energy sources or energy storage solutions to maintain a consistent energy supply. The ranking of countries in descending order (from the highest to the lowest PV power production) was Mali, Chad, Madagascar, Niger, Senegal, Burkina Faso, Central African Republic, Benin, Democratic Republic of Congo, Cameroon, Guinea-Bissau, Guinea, Togo, Ivory Coast, Congo Republic, Rwanda, Burundi, Equatorial Guinea, and Gabon. It is important to mention that the grid purchases represented the maximum power that could be purchased from the grid while the grid sales are considered as the maximum power that could be sold back to the grid. Concerning the grid sales, they

were between 11,604 (in Gabon) and 29,608 kWh/year (in Mali). The observation simply justified that with the highest PV power production, Mali was the country with the highest grid sales power production. The conclusions drawn from the power productions and the cost of energy are in good agreement in the context of sustainable energy transition since an available and affordable electricity was generated from the hybrid system.

### 3.5. Photovoltaic penetration

The photovoltaic penetration was defined as the ratio of PV electrical power output in a time step to the total electrical load served in this time step. Tables 10–28 and Fig. 10 show the PV penetration in each country. It was observed that it varied between 54.2 % (in Gabon) and 84.7 % (in Mali) with an average value of 71.65 %. Indeed, Mali was followed by Chad, Niger, Madagascar, Senegal, Burkina-Faso, Central African Republic, Benin, Democratic Republic of Congo, Cameroon, Guinea-Bissau, Togo, Ivory Coast, Congo Republic, Burundi, Rwanda, Equatorial Guinea, and Gabon. The highest PV penetration was found in South East Africa, followed by West Africa, East Africa, and Central Africa. The results are in perfect agreement with the PV power output.

## 4. Conclusion

The present paper investigated a techno-economic analysis of a Grid/Fuel Cell/PV/Electrolyzer hybrid system for hydrogen and electricity generation in some communities in the nineteen countries of the African and Malagasy Council for Higher Education (Benin, Burkina Faso, Burundi, Cameroon, Central African Republic, Congo Republic, Ivory Coast, Gabon, Guinea, Guinea-Bissau, Equatorial Guinea, Madagascar, Mali, Niger, Democratic Republic of Congo, Rwanda, Senegal, Chad, and Togo) in a strategy of sustainable energy transition. The novelty of the paper was threefold: (i) The Grid/Fuel Cell/PV/Electrolyzer hybrid system was modeled, simulated, and optimized in each of the aforementioned countries for the very first time; (ii) The levelized costs of hydrogen and energy were evaluated and compared; (iii) PV, fuel cell, grid purchases, and grid sales power productions were determined and compared. HOMER software was utilized for computations. The following conclusions were drawn.

- The minimum and maximum LCOE were US\$ 0.238/kWh (in Madagascar) and US\$ 0.344/kWh (in Gabon) respectively;
- The levelized cost of energy was the highest in East Africa (about US\$ 0.321/kWh) followed by Central Africa (about US\$ 0.288/kWh), West Africa (about US\$ 0.268/kWh), and South East Africa (about US\$ 0.238/kWh);
- The levelized cost of Hydrogen was the highest in East Africa (about US\$ 2.13/kg) followed by Central Africa (about US\$ 2.00/kg), West Africa (about US\$ 1.96/kg), and South East Africa (about US\$ 1.84/kg);
- The NPC was the highest in East Africa (about 210,920 US\$) followed by Central Africa (about 199,392 US\$), West Africa (about 195,637 US\$), and South East Africa (about 183,536 US\$);
- The highest PV penetration was found in South East Africa, followed by West Africa, East Africa, and Central Africa.

From all the results obtained, it was concluded that the Grid/Fuel Cell/PV/Electrolyzer hybrid system could be a promising solution in developing countries in general, and in the countries of the African and Malagasy Council for Higher Education in particular in the context of renewable and sustainable energy transition.

Some policy recommendations which could be formulated for energy transition in the African and Malagasy Council for Higher Education are.

- National policy in each of the aforementioned countries should offer a trajectory for the long-term energy mix which must contain a target

for renewable energy systems incorporating hydrogen as the energy storage technology;

- Economic incentives and financial policy instruments, similar to exemptions of the VAT, import, and income tax will decrease the upfront cost of the different components of the present architecture.

In the present work, different electrical load profiles (medium and high consumers) were not taken into account. In addition, various power sources (wind, biogas, geothermal for instance) were not considered in the hybrid system. The use of different technologies of storage (super-capacitor, and batteries for instance) was not also considered in the present architecture. Finally, different optimization algorithms were not implemented. All these limitations were left for future investigation.

### Credit author statement

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### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

### Data availability

Data will be made available on request.

### References

- [1] Cames. <https://www.lecames.org/>, 2023.
- [2] M.M.V. Cantarero, Of renewable energy, energy democracy, and sustainable development: a roadmap to accelerate the energy transition in developing countries, *Energy Res. Social Sci.* 70 (2020) 101716.
- [3] J. Kenfack, U. Nzotcha, J. Voufo, P.S. Ngohe-Ekam, J.C. Nsangou, B. Bignom, Cameroon's hydropower potential and development under the vision of Central Africa power pool (CAPP): a review, *Renew. Sustain. Energy Rev.* 151 (2021) 111596.
- [4] P.A. Østergaard, N. Duic, Y. Noorollahi, S.A. Kalogirou, Recent advances in renewable energy technology for the energy transition, *Renew. Energy* 179 (2021) 877–884.
- [5] O. Gershon, J.K. Asafo, A. Nyarko-Asamani, E.F. Koranteng, Investigating the nexus of energy consumption, economic growth and carbon emissions in selected african countries, *Energy Strategy Rev.* 51 (2024) 101269.
- [6] B. Khavari, A. Sahilberg, W. Usher, A. Korkovelos, F.F. Nerini, The effects of population aggregation in geospatial electrification planning, *Energy Strategy Rev.* 38 (2021) 100752.
- [7] S. Brauner, A. Lahnaoui, S. Agbo, S. Bösch, W. Kuckshinrichs, Towards green hydrogen? A comparison of German and African visions and expectations in the context of the H2Atlas-Africa project, *Energy Strategy Rev.* 50 (2023) 101204.
- [8] The world bank. <https://data.worldbank.org/indicator/EG.ELC.ACCS.ZS?locations=C9>, 2024.
- [9] O.D.T. Odou, R. Bhandari, R. Adamou, Hybrid off-grid renewable power system for sustainable rural electrification in Benin, *Renew. Energy* 145 (2020) 1266–1279.
- [10] J.H.R. Mensah, A.T.Y.L. Silva, I.F.S. dos Santos, N. de Souza Ribeiro, M. J. Gbedjinou, V.G. Nago, R.M. Barros, Assessment of electricity generation from biogas in Benin from energy and economic viability perspectives, *Renew. Energy* 163 (2021) 613–624.
- [11] R. Akpahou, F. Odoi-Yorke, L.K. Osei, Techno-economic analysis of a utility-scale grid-tied solar photovoltaic system in Benin republic, *Cleaner Engineering and Technology* 13 (2023) 100633.
- [12] D. Yamegueu, Y. Azoumah, X. Py, N. Zongo, Experimental study of electricity generation by Solar PV/diesel hybrid systems without battery storage for off-grid areas, *Renew. Energy* 36 (6) (2011) 1780–1787.
- [13] H. Hoff, M. Ogeya, D. de Condappa, R.J. Brecha, M.A.D. Larsen, K. Halsnaes, S. Liersch, Stakeholder-guided, model-based scenarios for a climate-and water-smart electricity transition in Ghana and Burkina Faso, *Energy Strategy Rev.* 49 (2023) 101149.
- [14] P. Tatsidjoudoung, M.H. Dabat, J. Blin, Insights into biofuel development in Burkina Faso: potential and strategies for sustainable energy policies, *Renew. Sustain. Energy Rev.* 16 (7) (2012) 5319–5330.
- [15] B.I. Ouedraogo, S. Kouame, Y. Azoumah, D. Yamegueu, Incentives for rural off grid electrification in Burkina Faso using LCOE, *Renew. Energy* 78 (2015) 573–582.
- [16] S. Baumert, A. Khamzina, P.L. Vlek, Greenhouse gas and energy balance of Jatropha biofuel production systems of Burkina Faso, *Energy for Sustainable Development* 42 (2018) 14–23.
- [17] M. Sawadogo, S.T. Tanoh, S. Sidibé, N. Kpai, I. Tankoano, Cleaner production in Burkina Faso: case study of fuel briquettes made from cashew industry waste, *J. Clean. Prod.* 195 (2018) 1047–1056.
- [18] B.I. Ouedraogo, D. Yamegueu, Techno-economic assessment of solar photovoltaic integration into national grids: a case study of Burkina Faso, *Energy Sci. Eng.* 7 (5) (2019) 1458–1468.
- [19] H. Abid, J. Thakur, D. Khatiwada, D. Bauner, Energy storage integration with solar PV for increased electricity access: a case study of Burkina Faso, *Energy* 230 (2021) 120656.
- [20] A. Ouedraogo, A. Diallo, S. Goro, W.D.A. Ilboudo, S. Madougou, D.J. Bathiebo, S. Kam, Analysis of the solar power plant efficiency installed in the premises of a hospital—case of the Pediatric Charles De Gaulle of Ouagadougou, *Sol. Energy* 241 (2022) 120–129.
- [21] E. Manirambona, S.M. Talai, S.K. Kimutai, A review of sustainable planning of Burundian energy sector in East Africa, *Energy Strategy Rev.* 43 (2022) 100927.
- [22] E. Manirambona, S.M. Talai, S.K. Kimutai, A review of sustainable planning of Burundian energy sector in East Africa, *Energy Strategy Rev.* 43 (2022) 100927.
- [23] E. Manirambona, P. Hendrick, T. Mushiri, Performance assessment and optimization of a micro-hydropower plant: case of Kigwena in Burundi, in: *AIUE Proceedings of the 18th Industrial and Commercial Use of Energy Conference*, 2020.
- [24] S. Godfrey, Redesigning a solar PV kiosk in high-temperature environments of Burundi, *Africa, Sustainability* 15 (6) (2023) 5514.
- [25] S. Bimenyimana, C. Wang, G.N.O. Asemota, A. Nduwamungu, C.K. Mesa, J. P. Ihirwe, Y. Mo, A techno-economic feasibility analysis for affordable energy system in the East African community countries, *Int. J. Photoenergy* (2021) 1–19.
- [26] J. Kenfack, F.P. Neirac, T.T. Tatietsé, D. Mayer, M. Fogue, A. Lejeune, Microhydro-PV-hybrid system: sizing a small hydro-PV-hybrid system for rural electrification in developing countries, *Renew. Energy* 34 (10) (2009) 2259–2263.
- [27] Y. Ayuketah, S. Gyamfi, F.A. Diawuo, A.S. Dagoumas, Power generation expansion pathways: a policy analysis of the Cameroon power system, *Energy Strategy Rev.* 44 (2022) 101004.
- [28] I.D. Ibrahim, Y. Hamam, Y. Alayli, T. Jamiru, E.R. Sadiku, W.K. Kupolati, A.A. Eze, A Review on Africa energy supply through renewable energy production: Nigeria, Cameroon, Ghana and South Africa as a case study, *Energy Strategy Rev.* 38 (2021) 100740.
- [29] E.G. Dountio, P. Meukam, D.L.P. Tchaptchet, L.E.O. Ango, A. Simo, Electricity generation technology options under the greenhouse gases mitigation scenario: case study of Cameroon, *Energy Strategy Rev.* 13 (2016) 191–211.
- [30] B.E.K. Nsafo, A.B. Owolabi, H.M. Butu, J.W. Roh, D. Suh, J.S. Huh, Optimization and sustainability analysis of PV/wind/diesel hybrid energy system for decentralized energy generation, *Energy Strategy Rev.* 32 (2020) 100570.
- [31] E.M. Nfah, J.M. Ngundam, Feasibility of pico-hydro and photovoltaic hybrid power systems for remote villages in Cameroon, *Renew. Energy* 34 (6) (2009) 1445–1450.
- [32] E. Muh, F. Tabet, Comparative analysis of hybrid renewable energy systems for off-grid applications in Southern Cameroons, *Renew. Energy* 135 (2019) 41–54.
- [33] N. Yimen, L. Monkam, D. Tchekam-Toko, B. Musa, R. Abang, L.F. Fombe, M. Dagbasi, Optimal design and sensitivity analysis of distributed biomass-based hybrid renewable energy systems for rural electrification: case study of different photovoltaic/wind/battery-integrated options in Babadam, northern Cameroon, *IET Renew. Power Gener.* 16 (14) (2022) 2939–2956.
- [34] R.H.T. Djela, P.T. Kapen, G. Tchuen, Techno-economic design and performance evaluation of Photovoltaic/Diesel/Batteries system through simulation of the energy flow using generated solar radiation data, *Energy Convers. Manag.* 248 (2021) 114772.
- [35] P.T. Kapen, B.A.M. Nouadje, V. Chegnimonhan, G. Tchuen, R. Tchinda, Techno-economic feasibility of a PV/battery/fuel cell/electrolyzer/biogas hybrid system for energy and hydrogen production in the far north region of Cameroon by using HOMER pro, *Energy Strategy Rev.* 44 (2022) 100988.
- [36] G.F. Tajouo, P.T. Kapen, F.L.D. Koffi, Techno-economic investigation of an environmentally friendly small-scale solar tracker-based PV/wind/Battery hybrid system for off-grid rural electrification in the mount bamboutos, Cameroon, *Energy Strategy Rev.* 48 (2023) 101107.
- [37] Y.W. Koholé, C.A.W. Ngouleu, F.C.V. Fohagui, G. Tchuen, Quantitative techno-economic comparison of a photovoltaic/wind hybrid power system with different energy storage technologies for electrification of three remote areas in Cameroon using Cuckoo search algorithm, *J. Energy Storage* 68 (2023) 107783.
- [38] C.A.W. Ngouleu, Y.W. Koholé, F.C.V. Fohagui, G. Tchuen, Techno-economic analysis and optimal sizing of a battery-based and hydrogen-based standalone photovoltaic/wind hybrid system for rural electrification in Cameroon based on meta-heuristic techniques, *Energy Convers. Manag.* 280 (2023) 116794.



- [39] M. Jahangiri, M.H. Soulouknga, F.K. Bardei, A.A. Shamsabadi, E.T. Akinlabi, S. M. Sichilalu, A. Mostafaeipour, Techno-economic optimal operation of grid-wind-solar electricity generation with hydrogen storage system for domestic scale, case study in Chad, *Int. J. Hydrogen Energy* 44 (54) (2019) 28613–28628.
- [40] M.H. Soulouknga, S.O. Oyedepo, S.Y. Doka, T.C. Kofane, Evaluation of the cost of producing wind-generated electricity in Chad, *International Journal of Energy and Environmental Engineering* 11 (2020) 275–287.
- [41] K. Bongole, Z. Sun, J. Yao, Potential for geothermal heat mining by analysis of the numerical simulation parameters in proposing enhanced geothermal system at Bongor Basin, Chad, *Simulat. Model. Pract. Theor.* 107 (2021) 102218.
- [42] A.I. Hassane, D.H. Didane, A.M. Tahir, J.M. Hauglustaine, B. Manshoor, M.F. M. Batcha, R.M. Mouangue, Techno-economic feasibility of a remote PV mini-grid electrification system for five localities in Chad, *Int. J. Sustain. Eng.* 15 (1) (2022) 177–191.
- [43] M.H. Soulouknga, T.E. Somefun, S.Y. Doka, Performance evaluation of wind turbines for sites in Chad, *Heliyon* 8 (11) (2022) e11565.
- [44] E. Kelly, B.A.M. Nouadje, R.H.T. Djiela, P.T. Kapen, G. Tchuén, R. Tchinda, Off grid PV/Diesel/Wind/Batteries energy system options for the electrification of isolated regions of Chad, *Heliyon* 9 (3) (2023) e13906.
- [45] I. Guaita-Pradas, I.M. Pérez, B.M. Soucase, Economical and financial study of photovoltaic energy production in Africa: case of Morocco, Côte d'Ivoire and South Africa, in: 2015 3rd International Renewable and Sustainable Energy Conference (IRSEC), 2015, pp. 1–6.
- [46] N. Khraief, O.E. Omoju, M. Shahbaz, Are fluctuations in electricity consumption per capita in Sub-Saharan Africa countries transitory or permanent? *Energy Strategy Rev.* 13 (2016) 86–96.
- [47] O.S. Ohunakin, O.J. Matthew, M.S. Adaramola, O.E. Atiba, D.S. Adelekan, O. O. Aluko, V.U. Ezekiel, Techno-economic assessment of offshore wind energy potential at selected sites in the Gulf of Guinea, *Energy Convers. Manag.* 288 (2023) 117110.
- [48] T.F. Ianda, A.D. Padula, Exploring the Brazilian experience to design and simulate the impacts of a biodiesel program for sub-Saharan countries: the case of Guinea-Bissau, *Energy Strategy Rev.* 32 (2020) 100547.
- [49] T.F. Ianda, A.D. Padula, Exploring the Brazilian experience to design and simulate the impacts of a biodiesel program for sub-Saharan countries: the case of Guinea-Bissau, *Energy Strategy Rev.* 32 (2020) 100547.
- [50] F.P. Garcia, A.K. Raji, Access to efficient and sustainable energy: case of Madagascar, in: 2020 IEEE PES/IAS PowerAfrica, 2020, pp. 1–5.
- [51] F.J. Andriamanohiarisoamanana, T.N. Randrianantoandro, H.F. Ranaivoarisoa, H. Kono, G. Yoshida, I. Ihara, K. Umetsu, Integration of biogas technology into livestock farming: study on farmers' willingness to pay for biogas in Madagascar, *Biomass Bioenergy* 164 (2022) 106557.
- [52] A.F. Touré, S.A. Addouche, F. Danioko, B. Diourte, A.E. Mhamedi, Hybrid systems optimization: application to hybrid systems photovoltaic connected to grid. A Mali case study, *Sustainability* 11 (8) (2019) 2356.
- [53] R. Bhandari, Green hydrogen production potential in West Africa—case of Niger, *Renew. Energy* 196 (2022) 800–811.
- [54] J.F.N. Touafio, O. Sanda, S. Malenguiza, J. M'Boliguipa, R.M. Mouangue, Analysis of a wind turbine project in the city of Bouar (Central African Republic), *Scientific African* 8 (2020) e00354.
- [55] K. Kusakana, H.J. Vermaak, Hybrid renewable power systems for mobile telephony base stations in developing countries, *Renew. Energy* 51 (2013) 419–425.
- [56] H.J. Vermaak, K. Kusakana, Design of a photovoltaic–wind charging station for small electric Tuk-tuk in DR Congo, *Renew. Energy* 67 (2014) 40–45.
- [57] E. Hurtado, E. Peñalvo-López, Á. Pérez-Navarro, C. Vargas, D. Alfonso, Optimization of a hybrid renewable system for high feasibility application in non-connected zones, *Appl. Energy* 155 (2015) 308–314.
- [58] I.K. Rice, H. Zhu, C. Zhang, A.R. Tapa, A hybrid photovoltaic/diesel system for off-grid applications in Lubumbashi, DR Congo: a HOMER pro modeling and optimization study, *Sustainability* 15 (10) (2023) 8162.
- [59] J.D.D. Niyonteze, F. Zou, G.N.O. Asemota, S. Bimenyimana, G. Shyirambere, Key technology development needs and applicability analysis of renewable energy hybrid technologies in off-grid areas for the Rwanda power sector, *Heliyon* 6 (1) (2020) e03300.
- [60] O. Bamisile, D. Cai, H. Adun, M. Taiwo, J. Li, Y. Hu, Q. Huang, Geothermal energy prospect for decarbonization, EWF nexus and energy poverty mitigation in East Africa; the role of hydrogen production, *Energy Strategy Rev.* 49 (2023) 101157.
- [61] de Dieu KH. Jean, S.P. Yoon, T.J. Kang, H.T. Kim, Y.S. Jeon, Y.C. Choi, Potential for peat-to-power usage in Rwanda and associated implications, *Energy Strategy Rev.* 13 (2016) 222–235.
- [62] E. Panos, M. Densing, K. Volkart, Access to electricity in the World Energy Council's global energy scenarios: an outlook for developing regions until 2030, *Energy Strategy Rev.* 9 (2016) 28–49.
- [63] J.A. Alzola, I. Vechiu, H. Camblong, M. Santos, M. Sall, G. Sow, Microgrids project, Part 2: design of an electrification kit with high content of renewable energy sources in Senegal, *Renew. Energy* 34 (10) (2009) 2151–2159.
- [64] F.A. Sarpong, J. Wang, B.B. Cobbinah, J.J. Makwetla, J. Chen, The drivers of energy efficiency improvement among nine selected West African countries: a two-stage DEA methodology, *Energy Strategy Rev.* 43 (2022) 100910.
- [65] S. Barakat, M.M. Samy, M.B. Eteiba, W.I. Wahba, Viability study of grid connected PV/Wind/Biomass hybrid energy system for a small village in Egypt, in: 2016 Eighteenth International Middle East Power Systems Conference (MEPCON), 2016, pp. 46–51.
- [66] N. Alshammari, M.M. Samy, J. Asumadu, Optimal economic analysis study for renewable energy systems to electrify remote region in Kingdom of Saudi Arabia, in: 2018 Twentieth International Middle East Power Systems Conference (MEPCON), 2018, pp. 1040–1045.
- [67] M.M. Samy, H.H. Sarhan, S. Barakat, S.A. Al-Ghamdi, A hybrid pv-biomass generation based micro-grid for the irrigation system of a major land reclamation project in kingdom of Saudi Arabia (ksa)-case study of albahra area, in: 2018 IEEE International Conference on Environment and Electrical Engineering and 2018 IEEE Industrial and Commercial Power Systems Europe (EEEIC/I&CPS Europe), 2018, pp. 1–8.
- [68] M.M. Samy, S. Barakat, Hybrid invasive weed optimization-particle swarm optimization algorithm for biomass/PV micro-grid power system, in: 2019 21st International Middle East Power Systems Conference (MEPCON), 2019, pp. 377–382.
- [69] M.B. Eteiba, S. Barakat, M.M. Samy, W.I. Wahba, Optimization of an off-grid PV/Biomass hybrid system with different battery technologies, *Sustain. Cities Soc.* 40 (2018) 713–727.
- [70] S. Barakat, M.M. Samy, M.B. Eteiba, W.I. Wahba, Feasibility study of grid connected PV-biomass integrated energy system in Egypt, *Int. J. Emerg. Elec. Power Syst.* 17 (5) (2016) 519–528.
- [71] A.F. Tazay, M.M. Samy, S. Barakat, A techno-economic feasibility analysis of an autonomous hybrid renewable energy sources for university building at Saudi Arabia, *Journal of Electrical Engineering & Technology* 15 (2020) 2519–2527.
- [72] M.M. Samy, M.I. Mosaad, M.F. El-Naggar, S. Barakat, Reliability support of undependable grid using green energy systems: economic study, *IEEE Access* 9 (2020) 14528–14539.
- [73] F. Odoi-Yorke, N. Abofra, F. Kemausuor, Decision-Making approach for evaluating suitable hybrid renewable energy system for SMEs in Ghana, *Int. J. Ambient Energy* 43 (1) (2022) 7513–7530.
- [74] C. Mokhtara, B. Negrou, N. Settou, B. Settou, M.M. Samy, Design optimization of off-grid Hybrid Renewable Energy Systems considering the effects of building energy performance and climate change: case study of Algeria, *Energy* 219 (2021) 119605.
- [75] A.F. Güven, N. Yörükeren, M.M. Samy, Design optimization of a stand-alone green energy system of university campus based on Jaya-Harmony Search and Ant Colony Optimization algorithms approaches, *Energy* 253 (2022) 124089.
- [76] F. Odoi-Yorke, J.J. Owusu, L. Atepor, Composite decision-making algorithms for optimisation of hybrid renewable energy systems: port of Takoradi as a case study, *Energy Rep.* 8 (2022) 2131–2150.
- [77] M.M. Samy, H.I. Elkhoully, S. Barakat, Multi-objective optimization of hybrid renewable energy system based on biomass and fuel cells, *Int. J. Energy Res.* 45 (6) (2021) 8214–8230.
- [78] M.M. Samy, S. Barakat, H.S. Ramadan, A flower pollination optimization algorithm for an off-grid PV-Fuel cell hybrid renewable system, *Int. J. Hydrogen Energy* 44 (4) (2019) 2141–2152.
- [79] F. Odoi-Yorke, A. Woenagnon, Techno-economic assessment of solar PV/fuel cell hybrid power system for telecom base stations in Ghana, *Cogent Engineering* 8 (1) (2021) 1911285.
- [80] S. Barakat, A. Emam, M.M. Samy, Investigating grid-connected green power systems' energy storage solutions in the event of frequent blackouts, *Energy Rep.* 8 (2022) 5177–5191.
- [81] M.M. Samy, A. Emam, E. Tag-Eldin, S. Barakat, Exploring energy storage methods for grid-connected clean power plants in case of repetitive outages, *J. Energy Storage* 54 (2022) 105307.
- [82] M.M. Samy, R.E. Almamlook, H.I. Elkhoully, S. Barakat, Decision-making and optimal design of green energy system based on statistical methods and artificial neural network approaches, *Sustain. Cities Soc.* 84 (2022) 104015.
- [83] A.F. Güven, M.M. Samy, Performance analysis of autonomous green energy system based on multi and hybrid metaheuristic optimization approaches, *Energy Convers. Manag.* 269 (2022) 116058.
- [84] S. Afonaa-Mensah, F. Odoi-Yorke, I.B. Majeed, Evaluating the impact of industrial loads on the performance of solar PV/diesel hybrid renewable energy systems for rural electrification in Ghana, *Energy Convers. Manag.* X 21 (2024) 100525.
- [85] L.K. Osei, F. Odoi-Yorke, R. Opoku, B. Baah, G.Y. Obeng, L.D. Mensah, F.K. Forson, Techno-economic viability of decentralised solar photovoltaic-based green hydrogen production for sustainable energy transition in Ghana, *Solar Compass* (2024) 100068.
- [86] F. Odoi-Yorke, S. Abaase, M. Zebilila, L. Atepor, Feasibility analysis of solar PV/biogas hybrid energy system for rural electrification in Ghana, *Cogent Engineering* 9 (1) (2022) 2034376.
- [87] [https://www.researchgate.net/figure/Flowchart-diagram-showing-the-simulati-on-process-of-HOMER-HOMER-evaluate-the-output-power\\_fig3\\_360270539](https://www.researchgate.net/figure/Flowchart-diagram-showing-the-simulati-on-process-of-HOMER-HOMER-evaluate-the-output-power_fig3_360270539), 2023.
- [88] Cames. <https://www.lecames.org/>, 2023.
- [89] Geographical coordinates finder. <https://www.geodatos.net/en/coordinates/>, 2023.
- [90] Map of Africa highlighting countries. [https://www.researchgate.net/figure/Map-of-Africa-highlighting-countries\\_fig1\\_332380173](https://www.researchgate.net/figure/Map-of-Africa-highlighting-countries_fig1_332380173), 2023.
- [91] <https://power.larc.nasa.gov/>, 2023.
- [92] <https://solargis.com/maps-and-gis-data/download/africa>, 2023.
- [93] J.A. Duffie, W.A. Beckman, *Solar Engineering of Thermal Processes*, second ed., Wiley, New York, NY, 1991.
- [94] [https://www.globalpetrolprices.com/electricity\\_prices/](https://www.globalpetrolprices.com/electricity_prices/), 2023.